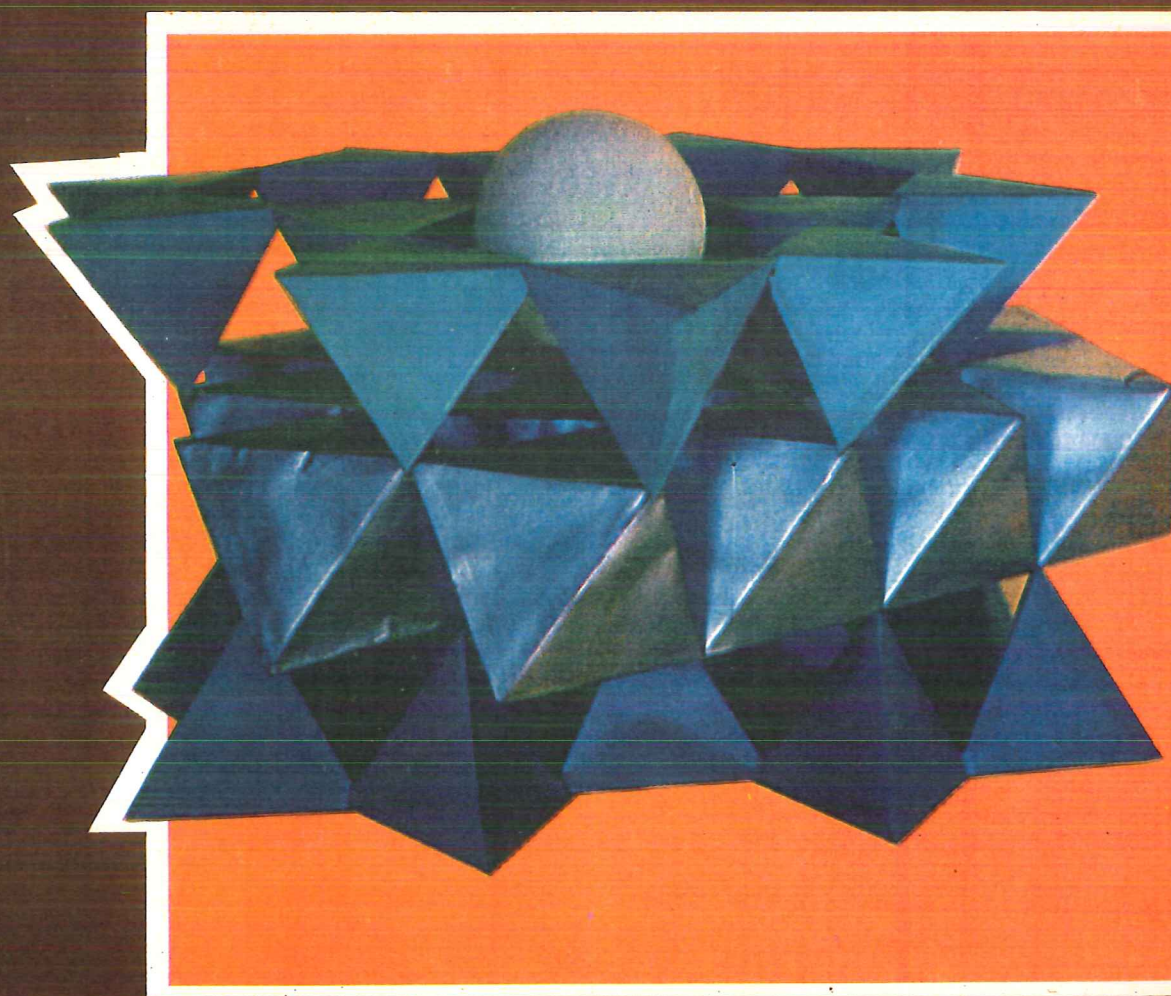


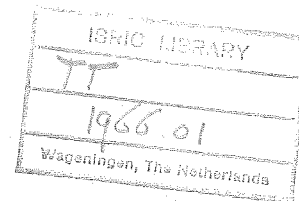


Land Capability Survey

OF TRINIDAD AND TOBAGO



No. 3 | NORTHERN TRINIDAD



LAND CAPABILITY SURVEY
OF
TRINIDAD AND TOBAGO

NO. 3

SOILS OF THE NORTHERN RANGE
OF
TRINIDAD

BY
C. B. BROWN and G. S. BALLY

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The Government of Trinidad and Tobago
The University of the West Indies
Texaco Oil Company
Shell Trinidad Limited
Tate and Lyle (Caroni Limited)
British Petroleum (Trinidad) Limited.

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Cover: Structural MODEL of a chemical compound found in soils. Model by C. B. Brown. Photography by Professor J. L. White of Purdue University, Indiana, U.S.A.

TABLE OF CONTENTS

	PAGE
SCOPE OF THE PROJECT	2
GENERAL DESCRIPTION	6
FACTORS INFLUENCING SOIL FORMATION	8
RELIEF AND PHYSIOGRAPHY	8
GEOLOGY	11
CLIMATE	15
VEGETATION	18
 SOILS	 21
METHODS OF STUDY	21
SOIL CLASSIFICATION	21
GENERAL DESCRIPTION OF SOILS	25
ACKNOWLEDGEMENTS	54
REFERENCES	55
APPENDICES	56
(1) SAMPLING PROCEEDURES AND ANALYTICAL DATA	57
(2) AREAS OF SOIL TYPES IN SLOPE CLASSES	90

LIST OF TABLES

I	AVERAGE TEMPERATURE IN TRINIDAD	15
II	RAINFALL IN THE NORTHERN RANGE.. .. .	16
III	SOIL CLASSIFICATION	23
IV	SUMMARY OF SOIL DATA	50

LIST OF FIGURES

1	LOCATION OF TRINIDAD	3
2	MAJOR PHYSIOGRAPHIC REGIONS	5
3	RELIEF AND DRAINAGE	9
4	GEOLOGY	12
5	ISOHYETAL MAP	14
6	MEAN MONTHLY RAINFALL	17
7	VEGETATION	19

INTRODUCTION

SCOPE OF THE PROJECT

The main purposes of this project are the survey of the soils of Trinidad and Tobago and the determination of their agricultural potential.

A wealth of information consisting of chemical and physical properties of soils, land capability, land use etc. has been collected and reported in a series of publications outlined below:

- Report No. 1 Land Capability Survey of Trinidad and Tobago:
Tobago — Soil Survey, Land Use, Land Capability
Irrigation Needs.
- Report No. 2 Profile Descriptions and Analytical Data for the
Soils of Tobago
- Report No. 3 Trinidad — Northern Range — Soil Survey,
Profile Descriptions and Analytical Data.
- Report No. 4 Trinidad — Central — Soil Survey, Profile
Descriptions and Analytical Data.
- Report No. 5 Trinidad — Southern — Soil Survey, Profile
Descriptions and Analytical Data.
- Report No. 6 Trinidad — Land Use and Land Capability.

This third report in the above series deals with the survey of soils of the Northern Range of Trinidad.

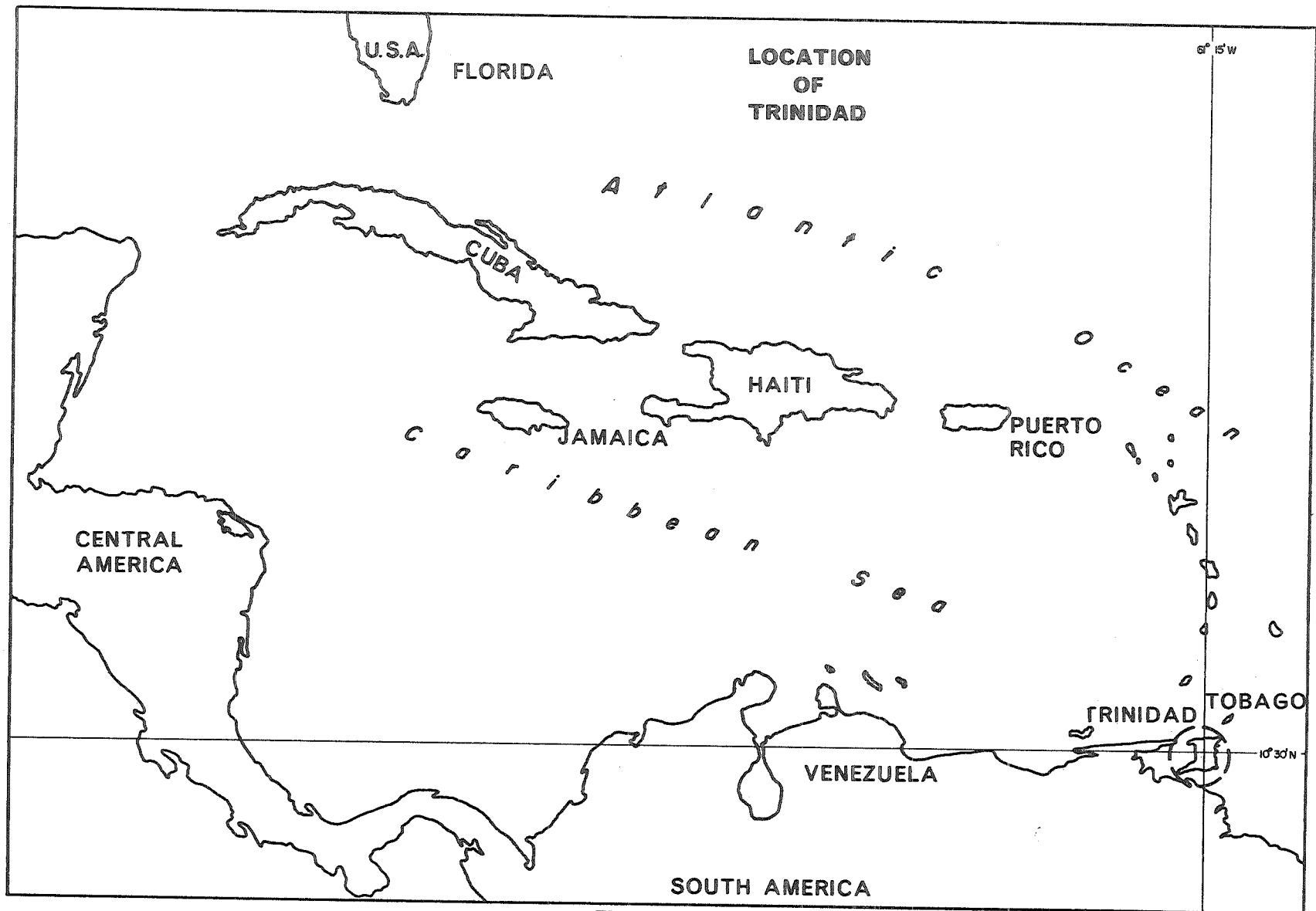


Figure 1

The Island of Trinidad was discovered in 1498 by Christopher Columbus on his third voyage.

Trinidad is the southernmost of the West Indian islands — an archipelago stretching from the tip of Florida, in the United States of America to the north eastern coast of Venezuela in South America. The island lies at 10° 30 ' north latitude and 61° 15 ' west longitude. The area is approximately 1,754 square miles.

Trinidad forms part of the independent nation of Trinidad and Tobago, having been granted independence by the United Kingdom in August 1963. It is a member of the British Commonwealth of Nations.

This report is concerned with approximately 420 square miles, generally known as the Northern Range.

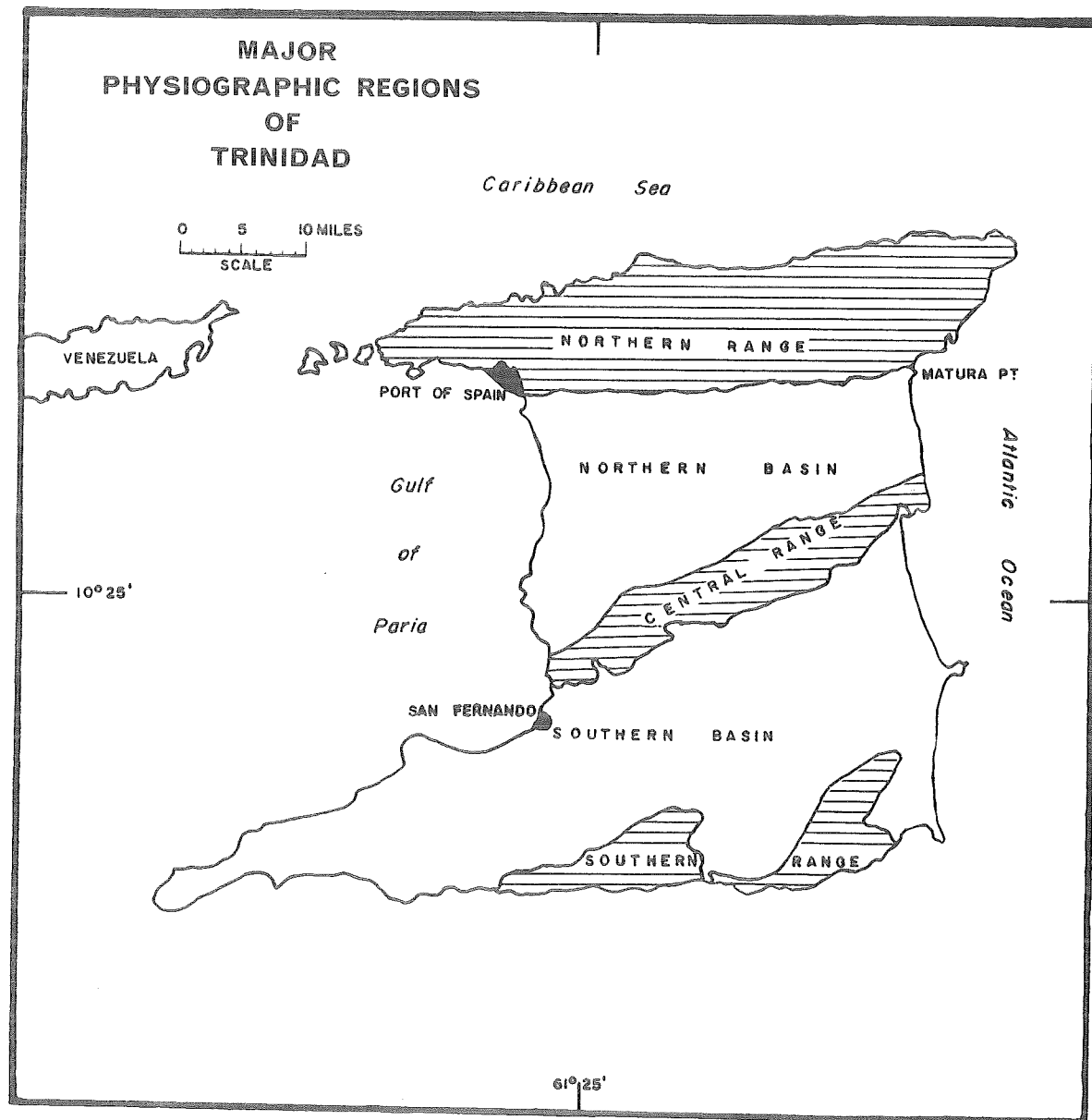


Figure 2

GENERAL DESCRIPTION

There are five physiographic regions in Trinidad. (Fig. 2). These have been defined by WARING (1926), SOUTHWELL (1928) and LIDDLE (1946), as follows:

- (a) The Northern Range
- (b) The Northern Basin
- (c) The Central Range
- (d) The Southern Basin
- (e) The Southern Range

The Northern Range covers that part of the island which lies north of a line drawn from Port-of-Spain on the west to Matura Point on the east coast. It is hilly, rugged country, almost inaccessible in some parts. This could explain why no detailed studies of the soils have ever been made. HARDY (1934) published a preliminary classification in which he recognised six soils, based on geology. WITT (1956) described eight series in a reconnaissance survey of the Range. In the present survey thirty-three individual soil series have been clearly defined. CHENERY (1952) described some of the soil types of the Northern Range. Previous work by the Soil Survey Division of the Ministry of Agriculture was made available to us.

The field work was carried out between May, 1963 and June, 1965. Our survey was detailed, in those sections which are easily accessible and are likely to be developed in the near future. Survey was reconnaissance in those sections which are difficult of access.

SOIL SURVEY

SOIL SURVEY

A: THE MAIN FACTORS INFLUENCING SOIL FORMATION

The end product of a land-capability survey is a map, accompanied by explanatory notes, on which are classified the land areas of the territory. On the land-capability map the soils are classified as to their inherent suitability or non-suitability for agriculture, pasture, forestry or other land use. It is axiomatic, and easily appreciated that before any land-capability classification can be stated, a knowledge of the soils is necessary. It may not however be as readily appreciated that a knowledge of the soils pre-supposes a thorough understanding of certain basic principles — the factors of soil formation.

The main natural factors which determine the types of soil that are found in the Northern Range are:—

1. *Relief and Physiography*
2. *Geology*
3. *Climate*
4. *Vegetation*

1: RELIEF AND PHYSIOGRAPHY

SUTER (1960) describes the Northern Range (Fig. 3) as a compact, hilly range with a single watershed. WARING (1926) considers the range as consisting essentially of two parallel series of ridges, of which the southern is somewhat the higher. The intervening lower lands (between the ridges) are drained via transverse valleys across the northern subsidiary ridge at a number of low points, e.g. the Marianne, Paria, and Madamas Rivers.

The southern or main ridge is however a continuous drainage divide with a comparatively low (1000-1500 feet) stretch, east of El Cerro del Aripo bisecting the Platanal — Cambural depression.

The long axis of the Northern Range is from east to west, and along this axis the Range is approximately sixty-miles: the average width is ten miles. Because of the mountainous terrain and consequent winding roadways, one may have to cover a distance of about 20 miles in order to cross the range from south to north. The distance from Port-of-Spain to North Post through Diego Martin is only 7 miles, but from Arima to Blanchisseuse, it is nearly 27 miles by driving road.



The average elevation of the Range is between 1500 and 2000 feet with the two highest peaks towering above 3000 feet — El Cerro Del Aripo 3085 and El Tucuche 3075. Other well known peaks are Chaguaramal 2817, Morne Bleu 2781, Pena De Juangalo 2395, El Cerro Del Oropouche 2159, Morne Poui 2003 and Zagaya 1226 feet.

The main divide lies about two-thirds of the way across the range, with the longer incline on the southern side, the land falling steeply towards the Caribbean Sea on the north. In several cases there is an almost sheer fall of over 500 feet from cliff to sea. North Post at the northern end of the Diego Martin Valley is 747 feet above sea level, and at Grande Matelot Point the elevation is 703 feet.

The natural drainage ways (Fig. 3) on the northern side of the Range — those which open to the Caribbean sea — are steep, narrow, gorge-like valleys. There are only five places on the North Coast where the rivers have any appreciable alluvial flats — Grande Riviere, Blanchisseuse, Fillette, Las Cuevas and Maracas. The associated rivers are Grande Riviere, Marianne River, Yarra River, Curaguat River and Burro River.

The southern or landward slopes are less steep and many of the larger rivers in Trinidad have their sources on the southern slopes of the Northern Range. The San Juan, St. Joseph, Tacarigua, Arouca, Oropuna, Guanapo, El Mamo, Aripo, Diego Martin, Chaguaramas and Tompire rivers have their beginnings in the Range. These, with the exception of the last three, join together to form the Caroni River — Trinidad's longest river — which meanders lazily across the Northern Basin, finally emptying into the Gulf of Paria on the west. The main drainage way on the east coast is the Oropouche River which receives as tributaries from the Northern Range — the Rio Grande, Melajo, Guaico and Cuare Rivers.

The valleys of the Range are generally transverse: the larger ones open to the south, and are the drainage basins of the rivers mentioned in preceeding paragraphs.

2 : GEOLOGY

The available geological data support the view that the Northern Range is an extension of the Coast Range of Venezuela and is part of the eastern branch of the Andean mountain system.

In so far as Agriculture in general, or Soil Science in particular is concerned, the geology of the Northern Range can be reduced to a relatively simple classification. There are three main groups of geological material (Fig. 4) which are easily recognizable viz.

- (a) Igneous Rocks
- (b) Metamorphic Rocks
- (c) Alluvial and Colluvial Deposits

Each of these groups constitutes the basic material for the formation of specific soil series. An understanding of this parent material aids considerably in the recognition of soil types.

(a) The Igneous Rocks

There is only one small area in Trinidad where igneous rocks are known to occur. This is at Sans Souci in the ward of Toco. Easily recognizable exposures are to be found along the coast between Sans Souci Point and Grande Riviere Point, where an igneous rock cliff about 100 feet high overlooks the sea. The igneous material reaches its maximum height of 903 feet above sea level at Manantial Hill in the ward of Toco.

The outcrop is generally massive and fine-grained, but weathers into large blocks. It is green to dark grey in colour and has been described as a basaltic rock of andesitic affinities. The rocks are commonly very deeply weathered and although portions may appear solid, they crumble easily under very little pressure. The joint-faces are often covered with thin clay films which are coloured black, but the internal mass of weathered rock is predominantly yellowish brown, light in weight and rather porous. The igneous rocks weather into a definite soil series.

(b) Metamorphic Rocks

In Trinidad, metamorphic rocks are found only in the Northern Range and they constitute the greatest portion of the geological strata of the region. There are several formations which are geologically separable, but are difficult to differentiate in the field e.g. the carbonaceous schists generally have calcareous associations. The four distinctive groups are:—

- (i) Micaceous schists
- (ii) Carbonaceous schists
- (iii) Calcareous schists and Limestones
- (iv) Schists with quartzitic grits

GEOLOGY OF THE NORTHERN RANGE

0 2 4 6 8 Miles

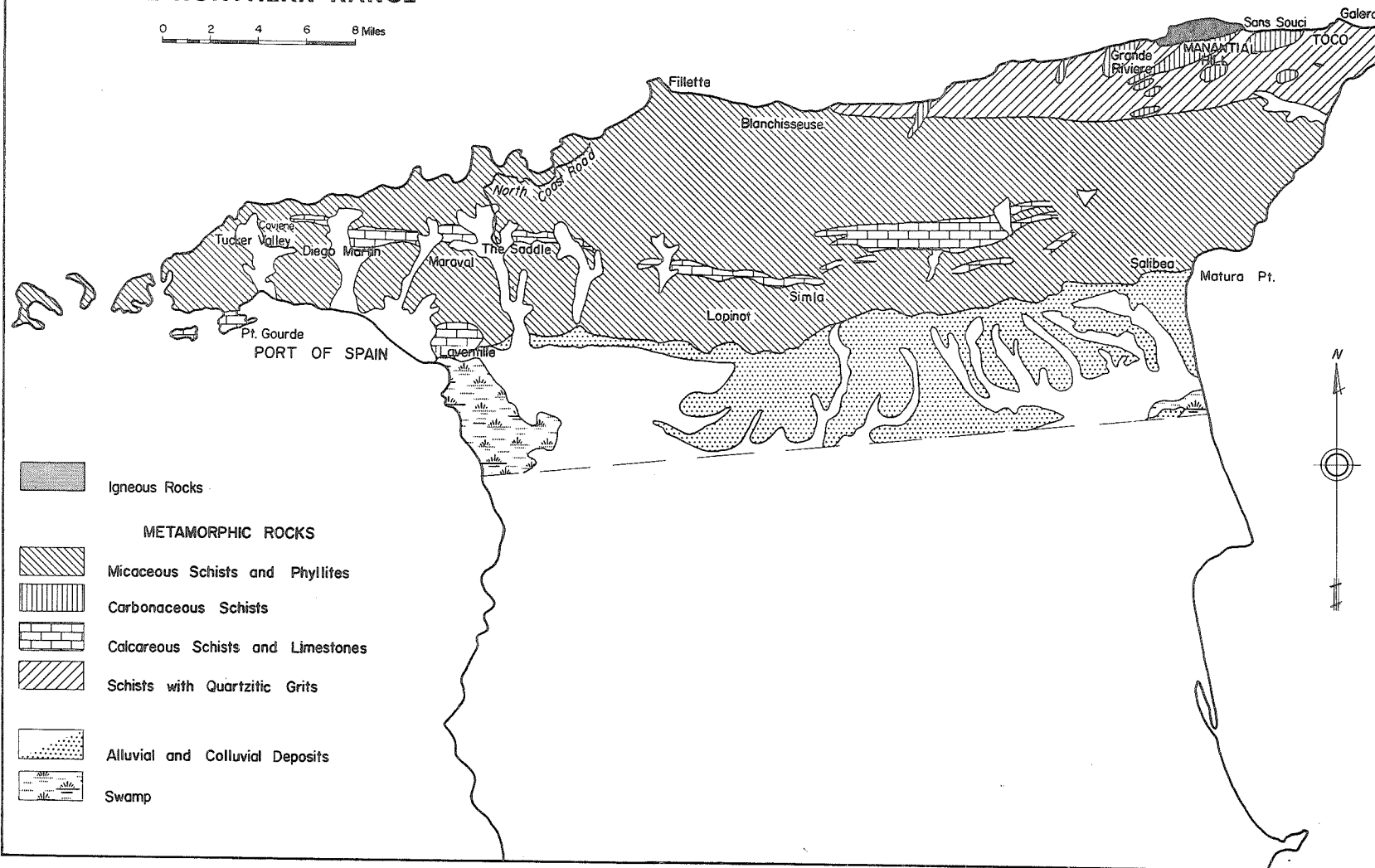


Figure 4

(i) *Micaceous Schists*

The largest portion of the Northern Range is of micaceous schist. SUTER (1960) considers the term phyllite to be more appropriate. These phyllites are easily recognized on any of the roads in the northern uplands of Trinidad. They are platy or laminated, strongly folded in parts and soapy to the feel; the colours are dark grey to black when fresh, but weathering to yellow or red with a silvery lustre. If SUTER'S nomenclature of phyllites be accepted, then it must be noted that there are more schistose types between Blanchisseuse and Fillette on the North Coast Road, and in the western parts of the Range. The common minerals are sericite, chlorite and quartz, with feldspars, pyrites and carbonaceous material as minor constituents. The dominant clay minerals, kaolin and illite, appear in the weathered derivatives as degradation products of sericite and chlorite.

(ii) *Carbonaceous schists*

These are black schists. Good examples of these occur on the Maracas Road. Laboratory examinations have shown that the black carbonaceous schists are themselves non-calcareous, but have veins of quartz and less commonly of calcite.

(iii) *Calcareous Schists and Limestones*

The calcareous schists are dark, soapy to the touch, and are high in mica content. They give the lithological background for the dark coloured soils, although some dark soils are found on the massive limestones.

Limestones are found at many places in the Northern Range, among them Laventille, Diego Martin, Lopinot and near "Simla" on the Blanchisseuse road in the quarries, as also at Maraval on the junction of the Saddle and North Coast Roads. The limestone deposits are known to be of different geological ages, but the soils that develop are quite independent of this. The massive limestone has a crystalline appearance, is dull grey in colour, and breaks with a pseudoconchoidal fracture. A dolomitic variety occurs at Point Gourde.

The massive limestones are extensively used as road metal, and are generally associated with the reddish soils of the Northern Range.

(iv) *Schists with Quartzitic Grits*

In several areas, especially around the Toco district, thick beds of quartzitic grits occur intercalated among the schist. These are generally very hard, coarsely crystalline rocks forming conspicuous coastal headlands as at Galera Point and along the north east coast, and prominent ridges inland. Finer grained beds approaching quartzites in type occur in the central and western part of the Range as at Blanchisseuse and Maracas.

In addition to the stratified rocks discussed above, injected quartz veins occur in many parts of the Range, often profusely. They are seen as conspicuous

ISOHYETAL MAP OF THE NORTHERN RANGE

0 2 4 6 8 Miles

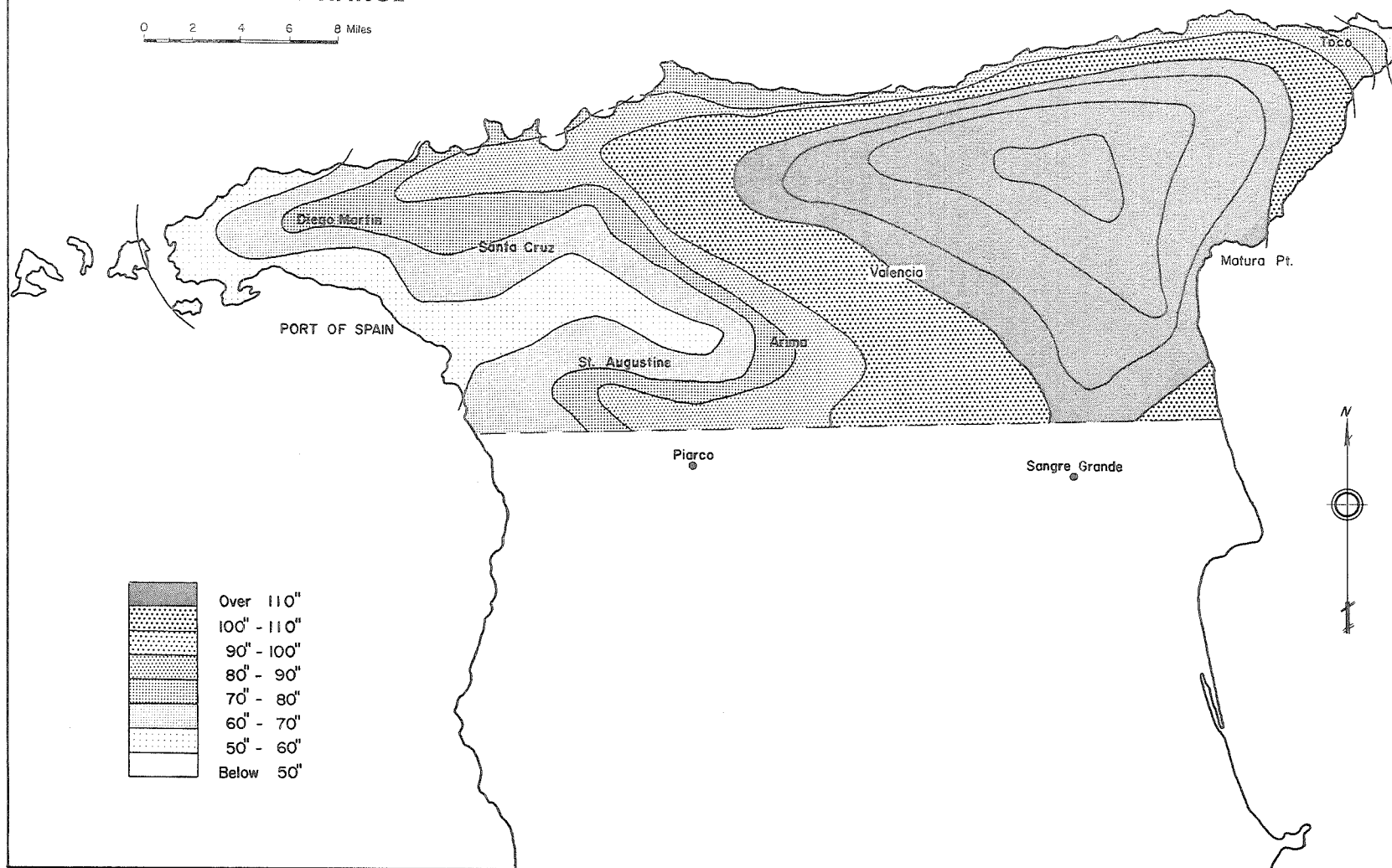


Figure 5

white veins cutting across the stratification of host rocks. They vary in thickness from less than one inch to several feet across. These quartz veins are important as the source of the quartz boulders and pebbles which make up the colluvial terraces and river gravels of the Northern Range rivers. The quartz boulders are remarkably large as for example on Covigne Road, Diego Martin, where one has a diameter of some five feet. Boulders two or three feet in diameter are common in most of the Northern Range rivers.

(c) Alluvial and Colluvial Deposits

Colluvial and recent alluvial deposits are found on the terraces, and in the valleys and flood plains of most of the northern rivers. Terrace deposits, apparently of recent geological age occur at Blanchisseuse, Galera and at several small outcrops on the Toco Main Road, east of Salibea. These deposits consist of coarse, generally rounded water-borne pebbles and boulders which vary from a gravel size to one foot in diameter, and constitute the parent material for definite soil series.

Diego Martin and Tucker Valley show good examples of recent alluvial deposits, and are surrounded on the landward sides by colluvia on the terraces.

3 : CLIMATE

The main climatic factors of rainfall and temperature influence the soils which develop in any area. The rainfall which reaches and passes through the soil carries with it, in solution varying amounts of soluble plant food. Obviously then, the greater the rainfall the deeper and more leached the soil is likely to be, so that the soils which occur in the higher rainfall belt of the Northern Range should be lower in plant nutrients than the drier soils. The temperature has a special effect in that the higher it is, the more easily will the plant nutrients become soluble. However, in so far as the soils of the Northern Range are concerned the climatic effect has been largely over-shadowed by the parent material and the topography. Geology with topo-sequence rather than climato-sequence is the key to an understanding of the soils of the Northern Range of Trinidad.

TABLE I
AVERAGE TEMPERATURE IN TRINIDAD

	TEMPERATURE FAHRENHEIT
Highest Maximum Temperature	95°
Lowest Minimum Temperature	60°
Average Maximum Temperature (Annual)	88°
Average Minimum Temperature (Annual)	72°
Maximum Diurnal variation	27°
Minimum Diurnal variation	5°
Mean Diurnal variation	17°

Source: *Government Meteorological Station, Piarco*

TABLE II

RAINFALL IN THE NORTHERN RANGE — 30 year mean rainfall, ending 1961

Eastern

STATIONS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEPT	OCT	NOV	DEC	TOTAL
Works & Hydraulics — Toco	3.30	2.96	1.72	2.70	4.78	8.55	7.93	8.16	6.28	7.55	8.20	6.90	69.03

Central

	JAN	FEB	MAR	APR	MAY	JUN	JULY	AUG	SEPT	OCT	NOV	DEC	TOTAL
Hollis Reservoir — Valencia	5.86	4.43	3.00	4.78	10.46	16.25	15.04	13.47	10.36	11.68	12.69	11.70	119.7
Torrecilla Estate — Arima	3.35	3.71	1.48	3.00	6.32	13.41	11.86	10.82	7.81	8.23	9.02	7.52	86.53
Sans Souci Estate — S. Grande	6.40	4.12	3.49	4.71	7.98	13.19	12.24	13.29	9.08	10.74	13.09	12.48	110.81
TOTAL													317.06
MEAN													195.68

	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEPT	OCT	NOV	DEC	TOTAL
River Estate — Diego Martin	3.76	2.77	2.11	2.38	4.36	7.95	9.92	11.09	8.45	7.57	7.25	6.54	74.15
La Pastora — Santa Cruz	5.48	4.59	3.05	2.20	3.23	8.52	10.60	13.12	9.16	6.49	7.48	7.10	81.02
St. Clair Expt. Station — P.O.S.	2.52	1.76	1.74	2.42	3.93	7.37	8.12	9.92	8.10	6.79	7.58	5.47	65.72
I.C.T.A. — St. Augustine	2.68	1.56	1.30	1.93	5.10	8.82	8.60	9.64	7.17	6.41	7.50	6.23	66.94
Meteorological Station — Piarco	2.86	1.67	1.36	2.20	5.23	9.44	8.93	9.72	6.88	6.20	7.71	6.65	68.85
TOTAL													356.68
MEAN													71.34

Source: Department of Works and Hydraulics

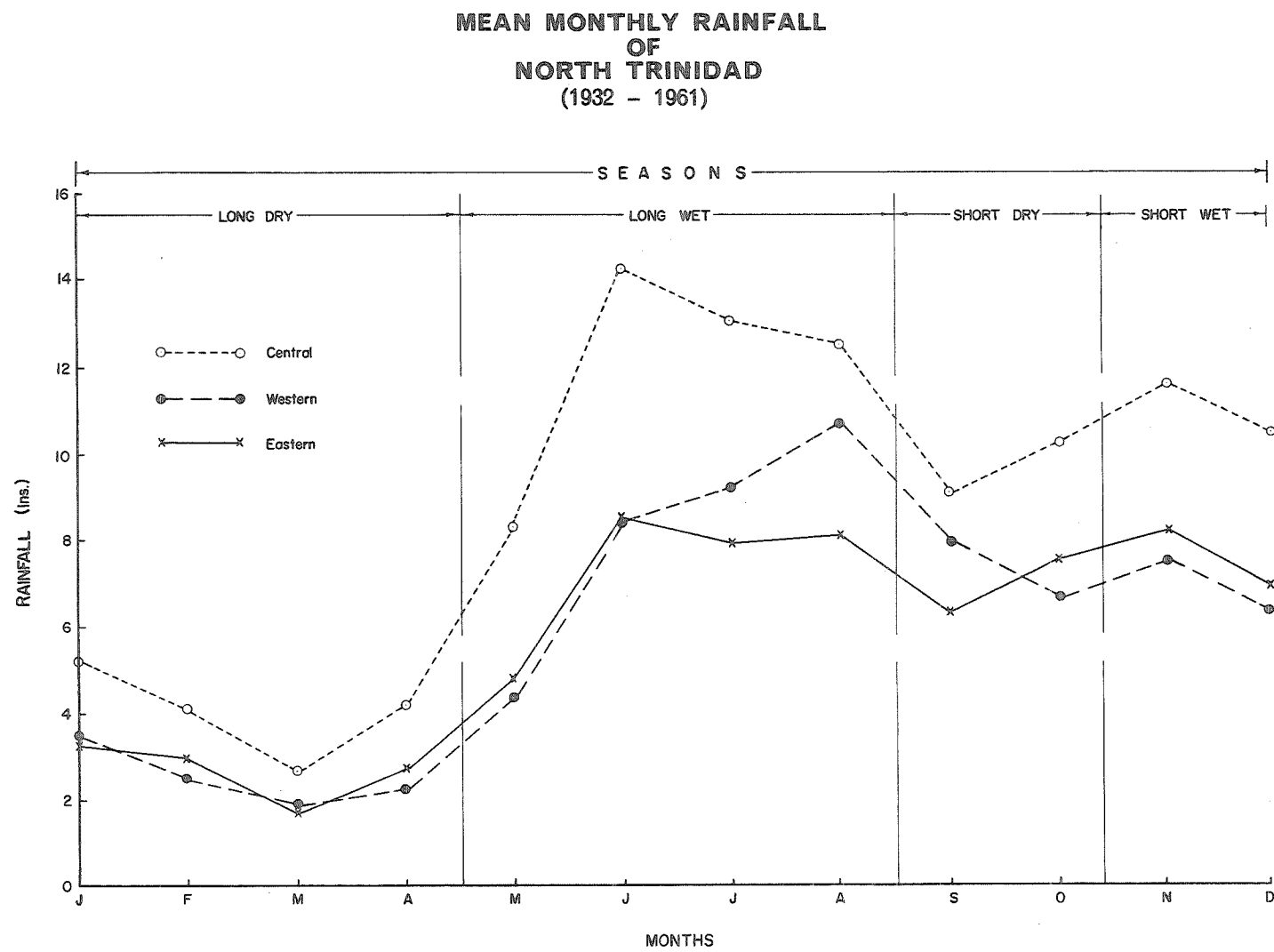


Figure 6

The island of Trinidad lies 10° north of the equator and the climate is therefore broadly tropical. The average annual maximum and minimum temperatures, and diurnal variations are given in Table I. These data have been kindly supplied by the Government Meteorological Station at the International Airport at Piarco. The climate is almost constant throughout the year — always hot and humid — but it is tempered by the breezes of the Atlantic.

There is a recognizable 'long rainy season' from May to September, and a short dry period locally called the 'petite careme' in September — October. The mean rainfall data for the dry and wet seasons are shown in Table II and Figures 5 and 6.

SMITH (1959) made a study of the potential evapo-transpiration and irrigation needs of Trinidad, in which he stated that the normal minimum required for most crops is 4 inches of water per month. Examination of Figure 6 shows that the average rainfall for February and March is less than 4 inches, with the outside months of January and April barely reaching the 4 inch mark. From an agronomic point of view, therefore, this is extremely important because water may become the limiting factor for plant growth during these months.

BEARD (1946) divided the Northern Range into three climatic regions supporting three types of natural vegetation — the lower montane, the montane and the upper montane forest. In our study of soils neither the various climatic regions nor the variation in rainfall had any extraordinary effect on the morphology of the soil, though we have found that in the heavy rainfall belt, (over 100 inches), the soils are somewhat deeper. In our opinion the depth is influenced more, by parent material and degree of slope than by the climatic conditions. In respect of parent material the reddish soils on limestone parent material, found in the Maraval area, are often deeper than three feet even on some very steep slopes. The soils on schists, on the other hand, vary with slope — the steeper the slope, the shallower the profile. Generally the topography and parent material have been more influential than any other factor of soil formation in the Northern Range, and consequently the soils are most easily studied as topo-sequences.

It is theoretically possible, that the high rainfall on the peaks of Aripo and El Tucuche may influence the profile, but agriculturally these are not significant areas. BEARD (1946) did observe that podsolization increased with height on Aripo.

4 : VEGETATION

There is an inter-relation between vegetation and soil formation, each one greatly influencing the other. The location and type of geological material initially determines the kind of vegetation that will succeed; this edaphic vegetation in turn influences the development of the profile. In a low lying flooded area, swamp grasses and mangrove will develop, and this type of indigenous vegetation will give rise to hydromorphic peaty soils.

NATURAL VEGETATION OF THE NORTHERN RANGE

0 2 4 6 8 Miles

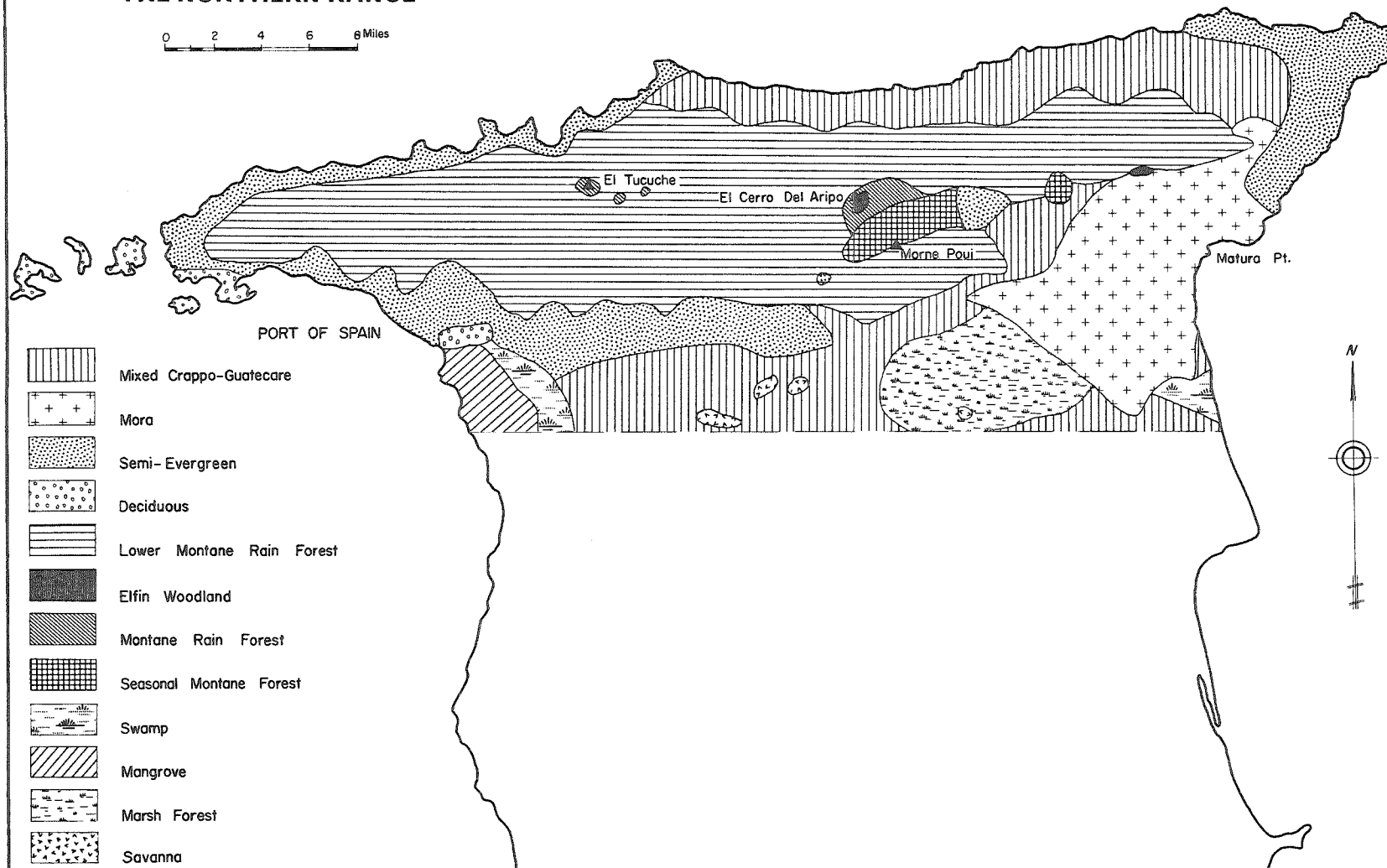


Figure 7

In the Northern Range, the virgin areas are covered with the climax vegetation — tropical rain forest. BEARD (1946) has shown a zonal distribution of climax forest vegetation varying from the wetter to the drier, and from the higher to the lower areas, Figure 7. In general though, natural vegetation has only influenced soils in the Northern Range in two ways:—

- (1) Under forest, the top soils of all series are darker.
- (2) Under forest, erosion is greatly reduced.

Many of the Northern Range valleys are planted with cocoa, and within the last fifty years, citrus and bananas have been introduced. The more accessible slopes, those near to main roads, have been used by peasant farmers for market garden crops under a system of shifting cultivation, which has both increased erosion and depleted the soil of its nutrients.

As a major factor of soil formation, the vegetation of the Northern Range has been over-shadowed by slope, parent material and to a lesser extent climate. Its initial lack of influence must not however be under-estimated, because any major removal of forest from the hills of the Northern Range will lead to destruction of the soils which are in existence now. The importance of vegetation as a major factor lies in the future. The forest must be maintained if the whole Northern Range is not to be eroded, leading into the more disastrous situation of flooding of the lowlands, removal of good topsoil, silting up of waterways — all of which add up to reduction in the agricultural potential of the territory.

B : THE SOILS

1 : METHODS OF STUDY

All the available data on the soils of the Northern Range were studied. Aerial photographs 1 : 28,000 and 1 : 25,000 were used as reference maps for 1 : 10,000 ward sheets. 1 : 10,000 maps were sufficiently large to take detailed information. Published maps are, for convenience in size, 1 : 25,000.

Soil pits were dug in each soil type for closer examination of the several horizons of the profile, and samples from each horizon were analysed both by the Central Analytical Laboratory of the University of the West Indies and the Soils Division of the Central Experiment Station of the Ministry of Agriculture.

Survey data were recorded on maps using the tripartite legend. A mapping symbol 91CI means soil series 91 which will be found to be Sans Souci loams with a C slope and erosion class I. An explanation of slope and erosion grouping is given below, while descriptions of the soils are given later.

Slope Classes

A	=	Slopes of dominant gradients.....	0°	—	2°
B	=	" " "	2°	—	5°
C	=	" " "	5°	—	10°
D	=	" " "	10°	—	20°
E	=	" " "	20°	—	30°
F	=	" " "	Greater than 30°		

Erosion Classes

- + Area of accretion — recent alluvial and colluvial deposits.
- 0 No apparent accretion or erosion.
- 1 Less than 50% of topsoil lost.
- 2 More than 50% of topsoil lost.
- 3 All topsoil and less than 50% of subsoil lost.
- 4 All topsoil and more than 50% of subsoil lost.
- 5 Eroding parent material.

2 : SOIL CLASSIFICATION

The whole of Trinidad has been considered as one mapping unit for the Land Capability Project. The whole unit has been divided into three sections for convenience of study viz:—

- (a) The Northern Range
- (b) Central Trinidad
- (c) South Trinidad

In each of these sections soils have been classified into three broad groups:—

- (i) Soils of the alluvial flats, plains and valleys.
- (ii) Soils of the terraces and subsidiary ranges.
- (iii) Soils of the uplands.

It will be easily appreciated that in dealing with any one section of the whole mapping unit, in the present case The Northern Range, that certain soils may not be found and this explains gaps in the expected consecutive numbering of soils in Table III. There are 33 known soils in the Northern Range. 62 in Central Trinidad and up to the present 20 have been described in the south. Some of these soils are common to more than one section so that in all, the Land Capability Survey has described 98 soil series to date.

In the classification list which follows there are two categories. In the higher category soils numbered 1 to 40 are lowland and alluvial soils; 41 to 60 are terrace soils; and upland soils carry numbers 61 to 99. In the sub-category, soils which are similar in lithology and drainage characteristics but differ in morphological, physical or chemical properties are grouped together by introducing a "hundreds" digit. Soil 23* is a freely drained lowland soil developed from hillwash: so also are 123 and 223 but they are morphologically separable. Another example of this system of classification is found in group 181, 281 and 381* which are upland soils similar to 81.

It has been considered important to signify where soils are calcareous and so all soils of free lime status are given symbol/L. Thus soil 24/L is by the convention used a lowland soil with high lime status. It is called the Grand Riviere series by the accepted convention of soil scientists to name the soil after the place at which it was first described.

Table III shows the soils of the Northern Range grouped according to this system. The missing numbers in the classification are either soils which do not occur in the Northern Range but exist elsewhere in the island, or soils which theoretically are likely to occur but have not been found in practice.

*Soils 23 and 381 do not occur in the Northern Range.

TABLE III
SOIL CLASSIFICATION

Group	Soils	Lithology	Drainage
Group A.	<i>Soils of the Alluvial Plains and Valleys.</i>		
A1.	Deep Cultivated Beach Sand with free internal drainage 1. Cocal Sand	Beach Sand	Free
A2.	Deep Hydromorphic Soil with restricted internal drainage 15.: Caroni Peaty Clay	Peat on clay	Impeded
A3.	Deep Alluvial Soils with free internal drainage 123.: Guanapo Series	Micaceous sand, gravel ..	Free
	223.: River Estate Series	Micaceous phyllites, sand ..	"
	323.: St. Joseph Series	Micaceous riverwash	"
	423.: Tacarigua Series	Micaceous loam	"
	24/L.: Grande Riviere Series	Mixed calcareous — carbonaceous schist alluvium.	"
A4.	Deep Alluvial Soils with restricted internal drainage 233.: Sangre Grande Series	Silty clay alluvium	Imperfect
	39.: Navet Clay	Clayey alluvium	Impeded
Group B.	<i>Soils of the Terraces and Subsidiary Ranges.</i>		
B1.	Terrace Soils with free internal drainage 41.: Blanchisseuse Series	Micaceous phyllites, quartz	Free
	141.: Galera sand	Quartzitic gravels and sands	"
	241.: Las Lomas Series	Levéé sand	"
	43.: Acono Series	Micaceous (bouldery) colluvia.	"
	143.: St. Augustine Series	Micaceous colluvia	"
	243.: Santa Cruz Series	Micaceous colluvia (with limestone association). ...	"
B2.	Terrace Soils with restricted internal drainage 53.: Anglais Series	Micaceous phyllites, quartz, sandstone.	
	153.: Cleaver Series	Loamy hillwash	"
	253.: Streatham Series	Micaceous sandy hillwash	Imperfect
	55.: Piarco Series	Sand and gravel	"
	255.: Valencia Series	Clayey sand, gravel	"
	59.: Long Stretch Series	Clay	"
Group C.	<i>Soils of the Uplands</i>		
C1.	Soil of the Intermediate Uplands with free internal drainage 261.: Arena Sand	Sand	Free
C2.	Soils of the Intermediate Uplands with restricted internal drainage 177.: Talparo Series	Clay	Impeded
	178.: Chickland Series	Shelly clay	"

		Lithology	Drainage
C3.	High Upland Soils with free internal drainage.		
	81.: Maracas Series	Micaceous phyllites ..	Free
	181.: Matelot Series	Micaceous phyllites ..	„
	281.: Spring Hill Series	Micaceous phyllites, talc	„
	82/L.: Diego Martin Series	Calcareous phyllites and limestone.	„
	182/L.: La Pastora Series	Limestone	Free
	282/L.: Maraval Series	Limestone	„
C4.	382/L.: Platanal	Limestone	„
	High Upland Soils internal drainage.		
	91.: Sans Souci Series	Igneous rock	Imperfect
	94/L.: Toco Series	Mixed Calcareous and carbonaceous phyllites ..	„

3 : GENERAL DESCRIPTION OF SOILS

Each of the thirty-two soil series is described below with respect to relief, parent material and morphological characteristics which include colour, texture and internal drainage conditions. A brief statement on present land use is given, followed by an assessment of the reaction and nutrient level of the soil. These descriptions are general, and full profile descriptions and analytical data are given in Appendix I.

GROUP A: SOILS OF THE ALLUVIAL PLAINS AND VALLEYS

SOIL NO. 1: COCAL FINE SAND

There is one type in this series. It is the only cultivated beach sand, and is found at Scotland Bay where it is under forest, at Las Cuevas, Maracas and several places along the seashore where it is under coconuts. Normally this type may be classified as a land-form, but since it is a good producer of coconuts, it must be considered as a soil type with economic agricultural potential.

Relief:

Flat.

Parent Material:

Beach Sand.

Profile Characteristics:

Under forest the topsoil is very dark brown with an organic mat. When exposed, it is yellowish brown and when dried it is whitish. This is a typical colour pattern of the highly siliceous soils in Trinidad. The subsoil is a concolorous light gray with dark brown stains due to deposition of organic matter in the profile. *CHENERY* (1952) defined this type as being uniform yellowish brown from surface to ground water table. While it does appear this way to the naked eye, the description given here fits the Munsell colours very closely.

The texture is sandy throughout the profile — fine sand averaging 70%, and coarse sand 10%.

Phase:

Deep.

Drainage:

Free to excessive internal drainage.

Present Land Use:

Coconuts and secondary forest.

Nutrient Status:

The entire profile is strongly acid and is deficient in all major plant nutrients, yet it supports coconuts well. A study of the nutrient level of the seepage water through the profile may help to explain why this apparently poor soil can support good stands of coconuts.

SOIL NO. 15: CARONI PEATY CLAY

The Caroni peaty clay is the only high humic hydromorphic soil of the Northern Range. It is derived from recent swamp deposits and is found along the coast, south of the Diego Martin Valley, and on the north coast at Las Cuevas and Maracas Bays.

Relief:

Flat.

Parent Material:

Recent swamp deposits.

Profile Characteristics:

The topsoil is dark brown to dark grayish brown humic clay with mangrove roots. The subsoil is dark gray clay with black to blackish brown peat, the peat content increasing with depth and at 2 feet the entire horizon is peat. Old Mangrove roots which are common in the profile are reddish brown in colour. The entire profile is saline, and inundated for the greater part of the year. In the height of the dry season, salt crusts form on the surface of the soil.

Phase:

Deep.

Drainage:

Impeded; inundated at high tide.

Present Land Use:

The salinity makes Caroni peaty clay naturally unsuitable for all normal agricultural crops. Mangrove, a tropical swamp indigene, flourishes well. Mangrove is used as fire-wood, and the bark is used in the tanning industry. Oysters which thrive under sheltered saline conditions are collected and sold as a delicacy. Where this soil has been drained and leached it has supported rice initially and sugar-cane after several years.

Nutrient Status:

The soil is very acid — pH below 5 — with high organic matter content. It is well supplied with the major nutrients — nitrogen, phosphorus and potassium — as well as magnesium and calcium but salinity and flooding are major limitations to its full utilization.

SOIL NO. 123 THE GUANAPO SERIES

The Guanapo loams are derived from alluvial and colluvial deposits of micaceous sand and gravel in the narrow trenches cut by the rivers of the Northern Range. These narrow flats must have been made by the rapid run-off from the rivers in the areas of high rainfall. This view is supported, firstly, by the observation that these soils only occur in the eastern (high rainfall) area of the mapping unit from Mausica eastwards; secondly, they are underlain by rounded to semi-rounded water-borne gravel. Cocoa had been the main crop grown on these soils previous to the Second World War, but since then vegetables have superseded cocoa. This is particularly so, on the Guanapo loams of Waller Field.

Relief:

Flat, in trenches or vegas on the banks of rivers and in mildly rolling country.

Parent Material:

Riverain, gravelly micaceous sand.

Profile Characteristics:

The Guanapo loams have a dark yellow or grayish brown topsoil which grades into various shades of brown or olive brown until a gravelly loam is encountered. The texture is sandy loam to loam throughout the profile until the gravelly loam is reached. Beneath this gravelly loam layer, beds of fine gravel-sized sandstone, slate, micaceous material and quartz are found. The entire profile has a high percentage of phyllite which imparts a smooth feel to the soil.

Phase:

Moderately deep.

Drainage:

Free to excessive.

Present Land Use:

Vegetables, cocoa, some citrus; gravel and sand mining for construction purposes.

Nutrient Status:

Both the acid and neutral members occur and both are low in nutrient content. The lowest horizons of the Guanapo loams are continuous with the river bottoms, so that although freely drained, they do not dry out completely during a normal dry season. Further, their proximity to the river makes irrigation relatively easy.

The neutral member occurs in areas where the river-wash comes from calcareous material such as Aripo and Platanal.

SOIL NO. 223: RIVER ESTATE SERIES

The River Estate loams are free drained soils developed from alluvium of the Northern Range. These soils are found in the broad valleys and river-basins surrounded by the rugged and hilly Northern Range e.g. Diego Martin and Santa Cruz. The profile always contains phyllitic fragments.

Relief:

Flat to very gentle slopes.

Parent Material:

Micaceous phyllite alluvium.

Profile Characteristics:

The topsoil is normally dark brown or dark yellowish brown. Below this, there occurs a yellowish brown horizon stained dark yellow brown from humic deposits along the dried root traces. The colour changes gradually to yellow and olive brown, streaked or spotted with black or red. The colour-changes are so gradual that often the profile appears concolorous. Small pieces of phyllite are recognised in the profile. The entire profile is silty to the touch, partly due to silt and partly due to phyllitic material brought down from the metamorphic rocks of the surrounding highlands. The texture distribution is sandy loam in the A horizon, becoming slightly heavier in the B and reverting to sandy loam in the C.

There are depressions in certain areas which cause a gleyed horizon to develop in the River Estate loams. Such soils have not been separately mapped. One known location occurs at Bellerande Road in Tucker Valley, and another is adjacent to the Western Main Road near Diego Martin. The blue gleyed horizon was found 2 feet in Bellerande Road and 6 feet in the Western Main Road pit. There are several other locations, among them being an area in Moka Estate, Maraval.

Drainage:

Internal drainage free; run off low.

Present Land Use:

Cocoa, coconuts, bananas, vegetables, pasture and housing.

Nutrient Status:

The pH is near neutral to slightly acid. Nitrogen is moderately good, phosphate and potassium are low, but calcium and magnesium are usually in moderate supply. The depression member, when near the seashore, may have a moderate content of sodium.

SOIL NO. 323: ST. JOSEPH SERIES

The St. Joseph series is developed on the recent riverain deposits of the area. They have been built by over-flow from the rivers and are usually 6 to 7 feet higher than the old river bed.

Relief:

Flat; flat pockets in otherwise steep terrain.

Parent Material:

Micaceous sand, colluvia and alluvia.

Profile Characteristics:

The profile is young and there are no definite signs of horizon development. The entire profile down to 6 feet is brown to olive in colour, with a slight darkening in the surface. The texture is sandy loam. The soil is slightly sticky, and contains bits of water-borne phyllite, quartz and sandstone. Gravel increases with depth from about 3 feet, and at 7 feet the entire horizon may be gravel and coarse sand.

Phase:

Deep.

Drainage:

Free to excessive.

Present Land Use:

Although the soil is very droughty, its close proximity to the river, with consequent ease of obtaining water makes it adaptable to green vegetables. Some citrus and cocoa are grown.

Nutrient Status:

The reaction is only moderately acid, with high phosphate, magnesium and calcium for a free-draining sandy soil. Nitrogen and potassium are low.

SOIL NO. 423: TACARIGUA SERIES

Acreage is less than 100.

SOIL NO. 24/L: GRANDE RIVIERE SERIES

The Grade Riviere series is an alluvial soil that is found in the flats surrounded by the high carbonaceous-calcareous hills. The most extensive area of this soil series is on the banks of the Grande Riviere. The entire profile is schistose with many micaceous fragments and it was noted that earthworms are usually present in the topsoil.

Relief:

Flat.

Parent Material:

Carbonaceous-calcareous schist alluvium.

Profile Characteristics:

The topsoil is dark gray in colour, with worm casts, and contains black carbonaceous schistose material. The subsoil is gray and olive with some marked dark gray splotches and, like the topsoil, contains black schistose particles.

The C horizon is sandstone and quartz gravel. The texture is sandy loam in the solum with an increase of gravel-sized particles in the C horizon. The effect of the dark coloured carbonaceous material is reflected throughout the entire profile.

Phase:

Deep.

Drainage:

Free.

Present Land Use:

Cocoa, vegetables and lastro or secondary bush.

Nutrient Status:

This series may range from moderately acid to neutral and is weakly leached. Nitrogen and potassium are only in moderate supply but phosphate is adequate in the rooting zone.

SOIL NO. 233: SANGRE GRANDE SERIES

This soil occurs in the alluvial flats of the rivers in the eastern section of the mapping unit. *CHENERY* (1952) limits its existence to the middle flood plains of the North Oropouche River system. He records that "it is derived from recent riverain silty clay which rests on the remains of an old mangrove swamp or lagoon". This soil series is liable to flood at least once in every three years.

Relief:

Flat.

Parent Material:

Silty clay alluvium.

Profile Characteristics:

The topsoil is yellowish brown, which becomes mottled orange at about 2 feet. There is a gradual change from yellowish brown to yellowish gray in the matrix colour, accompanied by a change in mottlings from orange to red. At about 6 feet depth, blue gley colours appear with manganese dioxide stains and orange and red mottlings, reflecting the imperfect drainage conditions of the profile. Peat is found at about 12 feet.

The entire profile is silty clay. Sand pockets which occur in the middle horizons, dry out and form semi-hard concretions.

Phase:

Deep.

Drainage:

Imperfect; intermittent flooding, although the water table may be always in the peat zone.

Present Land Use:

All the normal agricultural crops are known to grow on Sangre Grande silty clay. Cocoa, and recently citrus are common.

Nutrient Status:

The profile is acid; nutrient status is variable being influenced by material deposited during flooding.

SOIL NO. 39: NAVET CLAY

The only site of this series in the Northern Range mapping unit is a very small area on the banks of the stream which empties in the bay at Salybia.

Relief:

Flat.

Parent Material:

Clay alluvium.

Profile Characteristics:

The topsoil is dark brown humic clay. *CHENERY* mentioned the presence of films and crusts of iron hydroxide on the surface, but the area of the Navet series in the Northern Range did not show any of these properties.

The B horizon is grayish brown mottled yellowish or reddish brown and the parent material is blue-gray clay.

Phase:

Deep.

Drainage:

Impeded.

Present Land Use:

Swamp grass.

Nutrient Status:

Navet clay is flooded during the wet season. It has a moderately acid reaction but is moderately high in all major nutrients. Whenever Navet clay is drained, rice and green vegetables have been grown successfully. Sodium may be found in the profile.

GROUP B: SOILS OF THE TERRACES AND SUBSIDIARY RANGES.

The **Terrace soils** may be easily understood and recognised if they are thought of, as three broad groups:—

THE UPPER TERRACE SOILS

43	Acono
143	St. Augustine
243	Santa Cruz
53	Anglais
153	Cleaver (Intermediate)

THE LOWER TERRACE SOILS

253	Streatham
55	Piarco
255	Valencia
59	Long Stretch

A MISCELLANEOUS GROUP

41	Blanchisseuse
141	Galera
241	Las Lomas

The Upper Terrace soils are on the rising foothills of the Northern Range, are moderately steep and follow a fairly simple pattern.

St Augustine is a conspicuously red series that occurs generally on the southern slopes of the Northern Range, and is only found on the lower elevations, rarely higher than 100 feet. It generally sits neatly on the flanks of the phyllitic uplands.

Above 100 feet and stretching to approximately 500 feet elevation, the bouldery yellow *Acono soils* are found, often on almost flat-topped ledges.

The *Santa Cruz series* may be found anywhere in the whole range up to 500 feet, but occurs only in pockets which are closely associated with limestone uplands. In some cases this soil series is completely surrounded by limestone uplands, but the soil series itself is conspicuously non-calcareous.

The *Anglais* is the counterpart of the Santa Cruz, being however not associated with limestone hills. It occurs in an enclosed or partially enclosed valley of lower hills within the high Northern Range. It is a sort of colluvial soil, and except in very few cases is confined to the higher rainfall (eastern) belt of the mapping unit. It is closely associated with the upland soils and can at times be very steep. The *Cleaver* is mixed hillcreep. It is a transition type between the upper and lower terrace soils and may be relatively steep.

The Lower Terrace soils are *Piarco*, *Valencia*, *Long Stretch* and *Streatham* sited south of the upper soils and are on very mildly undulating or rolling country. They all have similar characteristics in respect of natural vegetation, present land use, and appearance of surface soil. These are the soils which occur on the lowest terraces of the Northern Range between St. Augustine and Balandra Bay. Externally they cannot be separated and as agricultural units they are essentially the same. It is only by means of internal morphological features that separations can be made, and these are dealt with in the individual descriptions.

Under the *Miscellaneous group* there is the *Las Lomas* which occurs generally on low hills—levées—two or three miles southwards from the lowest terrace of the Northern Range. The series has yellow sandy topsoils, over red B horizons and golden yellow C horizon, and is easily identifiable as low hillocks.

The *Galera* and *Blanchisseuse* series occur further east than any of the others, and form a broken arc from Balandra through Toco on to Blanchisseuse on the north coast. Both these soils may be moderately steep.

SOIL NO. 41: BLANCHISSEUSE SERIES

The Blanchisseuse series is an upland group of loams found on the subsidiary ridges of the Northern Range. These soils are situated on high undulating and sloping hills, and are confined to regions of high rainfall which are also exposed to the sea. The soils are derived from phyllites high in quartz content, and can be seen on the seaward slopes in Blanchisseuse. They extend as a broken arc from Blanchisseuse through Toco to Balandra.

The main agricultural crop is coconuts. It has been observed that there is normally a thick undergrowth containing a high proportion of prickly-edged grasses.

Relief:

Undulating with moderate slopes and exposed to the sea.

Parent Material:

Phyllites and schists with high quartz content.

Profile Characteristics:

This is a shallow to moderately deep soil having a dark brown top with many roots. The dominantly gray subsoil has various shades of reddish yellow and reds, and is always stony containing quartz, sandstone and schist remnants. The schist of the subsoil is either yellow or olive, but the olive is not associated

with any form of restricted drainage. This horizon grades into the quartzitic schist parent material, into which and through which roots pass very easily.

Phase:

Shallow to moderately deep.

Drainage:

Free.

Present Land Use:

Mostly coconuts, forest, tonca beans and rarely cocoa.

Nutrient Status:

The soil is strongly acid throughout the profile, moderate in all the major nutrients and weakly leached, except in the surface horizon. The abnormally high base saturation in the surface horizon may be due to high calcium and magnesium held by the organic colloids formed from leaf litter. It is not uncommon to find sodium in this soil.

SOIL NO. 141: GALERA SERIES

This soil series consists of two easily identifiable types — coarse sandy loam and coarse sand — and is very extensive between Balandra and Toco. It is found on the terraces bordering the sea, in the normal topo-sequence at an elevation just below the Blanchisseuse series. The position could be described as a raised beach associated with terrace deposits of recent geological age.

Relief:

Undulating terraces.

Parent Material:

Gravelly sand

Profile Characteristics:

The topsoil is sandy loam or sand with an abundance of roots and is usually dark grayish brown when wet, and grayish white when dry. The subsoil is a dark yellow brown and brown sandy loam with a high percentage of gravel. The parent material is a gravelly loam stained light red and yellow brown. In the higher areas the subsoil is lighter in colour.

The entire profile is sandy. Small, almost perfectly spherical gravel usually less than $\frac{1}{2}$ inch diameter appears at about 3 feet and continues downwards until the parent material is reached. Water seeps slowly through the soils of the series. This seepage is due to the intermediate topographic position of the soils.

Phase:

Moderately deep.

Drainage:

Free with water seepage.

Present Land Use:

Mostly coconuts; some limes, few vegetable gardens.

Nutrient Status:

The series is very acid with moderate nitrogen, potassium, calcium and magnesium but low phosphate. A moderate amount of sodium is found in the surface and this may be due to sea-spray. The sand content averages about 70%

in the profile, but unlike the Cocal series, 60% of the sand is coarse. In the Cocal series 70% of the sample is fine sand. The soil is droughty and young coconuts suffer from lack of water during the long dry season, as evidenced by the scorched appearance of the older leaves.

SOIL NO. 43: ACONO SERIES

The Acono Series is also found on terraces of the Northern Range and is invariably confined to the narrow valleys. The terrace usually occurs between the 100 and 500 foot contour. This series can be considered to be the higher elevation counterpart of the St. Augustine series. Acono soil has been cultivated in several common crops, but it appears that its productivity is good, only when strict management systems are practised.

There is a great similarity between the colour profile of the Santa Cruz and the Acono series. They differ however in that the Acono is normally at a higher contour, is found in association with phyllitic uplands and appears generally less productive. A further difference is that the Santa Cruz series has limestone associations. Analytical data show some differences in that the Acono is more acid and has less available nutrient ions than the Santa Cruz.

Relief:

Sloping to nearly flat terraces.

Parent Material:

Micaceous phyllites, quartz and sandstone.

Profile Characteristics:

The topsoil is usually dark brown, but in open eroded areas this dark soil is missing, and a yellow brown horizon is exposed at the surface. Under this comes an olive brown and yellow brown layer which is mottled various shades of red. The subsoil consists of an intermixture of various shades of yellow and red. Quartz boulders, up to 2 feet in diameter, sandstone and weathering phyllite may be seen at any depth in the profile. It has been found that in spite of this stoniness, root penetration is very good.

Texture profile is normal — sandy loam becoming heavier in the B horizon.

Phase:

Deep, often severely eroded.

Drainage:

Free.

Present Land Use:

Tonca beans, cocoa, vegetables and secondary forest.

Nutrient Status:

Strongly acid, low in all nutrients and very strongly leached.

SOIL NO. 143: ST. AUGUSTINE SERIES

The St. Augustine series is one of the several terrace soils of the Northern Range. In the broader valleys it normally occurs between the bottom land soil — the River Estate — and the upland soils. The profile is conspicuously red but

may have a yellowish background with a fairly high distribution of quartz, and it usually occurs below the 100 foot contour.

Relief:

Sloping terraces of the Northern Range.

Parent Material:

Detritus of phyllites and quartz.

Profile Characteristics:

The topsoil is dark yellowish brown with a red cast, but may be masked to grayish brown by organic matter. The subsoil is reddish brown and the red colour increases with depth until the C horizon is reached. The C horizon is conspicuously red with yellow mottlings. Quartz and silvery phyllite fragments occur at any depth in the profile. The entire profile is in the loam class although there may be minor textural variations.

Phase:

Deep, stony.

Drainage:

Internal drainage free; run-off moderate.

Present Land Use.

Scrub as a result of shifting cultivation, coconuts mixed with vegetables, pastures and housing. Port-of-Spain, old Diego Martin and the villages from Laventille to San Juan are built, for the most part, on St. Augustine loams.

Nutrient Status:

The series has a good tilth, is very acid and low in all the nutrients except nitrogen which is moderate.

SOIL NO. 243: SANTA CRUZ SERIES

The Santa Cruz series is developed from colluvial material on terraces of the Northern Range, and is usually on the foothills of the limestone areas. The lower horizons are stony and rock fragments may be found in the surface of the soil in the shallow phase. Plant growth on these terraces is good. This may be due to the fact that it is the least acid of all the terrace soils, and is only moderately leached.

Relief:

Sloping terraces to nearly level land.

Parent Material:

Colluvial material of the Northern Range — transported and partly weathered metamorphic phyllites in association with quartz and sandstone.

Profile Characteristics:

The topsoil in the shallow phase is dark brown. Below this, is a brown or yellow horizon which takes on a reddish cast below 10 inches. In the reddish horizon there is a high concentration of transported rocks of phyllite and sandstone. The shallow Santa Cruz soil may vary from 12 to 30 inches, and generally occurs nearer the hills than the deep phase. Thus, it is not uncommon to find calcareous chips within the profile.

The deep profile has a similar colour configuration except that the bright reddish coloured horizon occurs at greater depth, about, 2 feet.

Texture is in the sandy loam class with a smooth feel which must be partly due to phyllite particles in the profile.

Phase:

Shallow, moderately deep, stony.

Drainage:

Internal drainage free; run off low.

Present Land Use:

Mostly cocoa and citrus, with small amounts of bananas, food forest, vegetables, limes, coconuts, mangoes and other fruit trees.

Nutrient Status:

Santa Cruz soil series is only moderately acid, and although moderately leached is very low in all the major nutrients.

In spite of high sand content and consequent free internal drainage these soils are not particularly droughty. An explanation is that they are usually found in sheltered valleys away from the drying winds and are also moistened by dew.

SOIL NO. 241: LAS LOMAS SERIES

These soils are commonly developed on the lower elevations of the Central Range. In the present mapping unit there are few small outcrops just south of the Northern Range generally on hillocks. Flat phases are also found.

Relief:

Generally low hills, occasionally flat.

Parent Material:

Terrace and levee sand.

Profile Characteristics:

The topsoil is brownish yellow which may be darkened by organic matter. At about 3 feet the brownish yellow colour is superseded by a strong yellowish gold colour with reddish and orange mottlings. The red and orange splotches increase in size and concentration, until at about 5 feet, the profile is bright and brilliant red. The C horizon reverts to strong yellow with weak orange mottlings. The A and B horizons may be loam to sandy loam, but the parent material is a sand.

Phase:

Deep, eroded when the red horizon is very near the surface.

Drainage:

Free.

Present Land Use:

Calcifuge crops, secondary forest or scrub, citrus, forest.

Nutrient Status:

The series is strongly acid, subject to desiccation in the dry season and is also low in all nutrients.

SOIL NO. 253: STREATHAM SERIES

Acreage is less than 100.

SOIL NO. 53: ANGLAIS SERIES

In the sheltered valleys of the high rainfall belt of the Northern Range this moderately deep productive soil series develops. It has been long recognised as being inherently productive. Wherever this soil series occurs, it is always used intensively for agriculture.

The main factors responsible for the high productivity of the series are adequate moisture, good depth and the sheltered position.

Relief:

Rolling and undulating sheltered hills, sometimes very steep.

Parent Material:

Micaceous colluvia, sandstone and some quartz.

Profile Characteristics:

The topsoil of the series varies from dark grayish brown to dark brown with a moderately high humus content. The second horizon is light olive brown and light yellowish brown with very fine red mottlings. Remnants of colluvia containing phyllites, sandstone and very little quartz are found in this horizon and the colluvial material increases with depth.

The texture is loamy in the surface but gets heavier with depth.

Phase:

Moderately deep.

Drainage:

Imperfect.

Present Land Use:

Cocoa, bananas, coffee, tonca-beans, forest.

Nutrient Status:

The soil is a loam, moderately leached, strongly acid but has moderately good supply of all nutrients except phosphate.

SOIL NO. 153: CLEAVER SERIES

These loams are found in the middle terrace class. They occur as a thin band along the southern foothills of the Northern Range with isolated patches on the northern flanks, one of which is near Las Cuevas Bay.

Relief:

Undulating foothills.

Parent Material:

Mixed detrital loamy hillwash.

Profile Characteristics:

This profile is comprised of a horizon of dark brown sandy clay loam or sand over an orange red to scarlet red sandy or silty clay with many stones or pebbles. The quantity and diameter-size of the stones increase with depth and

at about 5 feet, individual quartz stones may be greater than a foot in diameter. The matrix is yellow to red.

The texture profile is sand to sandy clay loam in the surface, gravelly sandy or silty clay in the B, gravelly to bouldery silty clay in the parent material.

Drainage:

Imperfect.

Present Land Use:

Shifting peasant gardens, scrub and forest.

Nutrient Status:

The soil is acid, strongly leached and low in all the major nutrients. The surface (3 inches) in the profile described in the appendix is moderately nutrient-rich, but this may be due to organic matter in a virgin profile.

SOIL NO. 55: PIARCO SERIES

This is the least widespread of all the lower terrace soils in the mapping unit, and like the other two lower terrace soils, is heavily "hogwallowed" in the surface.

Relief:

Mildly undulating.

Parent Material:

Terrace sand and gravel on clay.

Profile Characteristics:

The topsoil is a grayish brown fine sandy loam under vegetative cover; when exposed it is whitish gray. Under this comes a whitish horizon which CHENERY (1952) describes as bleached. The B horizon is gray, initially mottled very sparsely with orange, weak yellow and red, but the red mottlings increase with depth, so that in the lower part of the B horizon the entire profile is splotted, streaked and mottled strongly and conspicuously red in the weak gray matrix. Below this comes the blue gleyed horizon which may have yellow or red splotches. Iron concretions are found in the solum.

The topsoil is sand to sandy loam followed by sand, and finally silty clay.

Phase:

Deep.

Drainage:

Imperfect; water-logged in the wet and desiccated in the dry season.

Present Land Use:

Sparse grasses, wild palms, marsh forest.

Nutrient Status:

The series is water-logged for most of the wet season, completely desiccated in the dry season, although the water table may be only a few feet below surface. The profile is extremely acid and it is almost devoid of any plant nutrients.

SOIL NO. 255: VALENCIA SERIES

The Valencia series is another of the terrace soils occurring below the 150 foot contour. The surface conditions are almost identical with the Piarco and

Long Stretch series. *CHENERY* describes it as his best example of a tropical ground-water podsol.

Relief:

Mildly undulating.

Parent Material:

Quaternary sands and gravelly clays.

Profile Characteristics:

As in the Long Stretch and the Piarco, the surface is "hogwallowed". There are 2 to 4 inches of forest litter. Under this is a horizon of fine quartz sand, stained brown from humus. When the vegetative cover is cleared and the soils dry out the sand grains are whitish gray — a common feature of siliceous sands. Under forest there is a bleached whitish A2 horizon of quartz and gravel. Next follows a layer of humus-cemented sand and gravel underlain by hard iron-cemented sand and gravel. The pan restricts internal drainage, and below the pan gleyed colours of whitish and bluish grays, both mottled orange, appear.

In respect of texture the upper horizons — above the cemented pan — are sandy and gravelly; then follow the clayey gravels and sandy clays, and finally the lower horizons which are gleyed sands, clays and gravels. In the wet season, "hogwallow" depressions are filled with water but become completely desiccated in the dry season, although the water table may only be a few feet down. Capillary action to the surface is restricted by the cemented pan in the profile.

Phase:

Deep.

Drainage:

Imperfect; water-logged in the wet and desiccated in the dry season.

Present Land Use:

Forest, shifting peasant gardens and secondary forest or scrub.

Nutrient Status:

In nutrient status, this series is very similar to the Long Stretch. This soil is very acid and lacking in all the nutrients, except for those which may be deposited on the surface from leaf-fall under forest cover. The Valencia responds to cropping in the same way as the Long Stretch, one or two crops of annuals then failure, but it does have a higher pH value than any of the lower terrace soils.

SOIL NO. 59: LONG STRETCH SERIES

The Long Stretch series is one of the group of soils which is developed on detrital terraces between the 50 and 150 foot (approximately) contours, on the southern foothills of the Northern Range. Large areas are found from Arima eastwards to Sangre Grande, generally south of the Eastern Main Road, although patches do exist north of the Road. These soils are usually under forest, as are the other lower terrace soils — Valencia and Piarco.

Relief:

Mildly undulating.

Parent Material:

Quaternary silty clay.

Profile Characteristics:

The description given here is very similar to that of *CHENERY* (1952). The surface is heavily "hogwallowed", with a humic mat. The true topsoil is a brownish gray over a yellow putty clay with marked orange and brown mottlings and specklings. The subsoil has red and yellow mottlings and is weakly cemented. A whitish horizon with red mottlings follows and under this, there is a blue clay with dark red to purple mottlings.

The A horizon may be silty clay or sandy clay overlying silty clay and finally clay in the subsoil.

Phase:

Deep, occasionally stony.

Drainage:

Impeded, water-logged in the wet and desiccated in the dry season.

Present Land Use:

Forest, shifting peasant gardens and secondary forest or scrub.

Nutrient Status:

The profile is extremely acid, is low in all nutrients although the organic mat may have an adequate supply of nitrogen, phosphorus, potassium. Thus, when the virgin Long Stretch is cleared and planted it gives one or two good crops of annuals — peas, corn or vegetables. When the organic complex is used up, the soils become unproductive and are abandoned.

GROUP C: THE UPLAND SOILS

SOIL NO. 261: ARENA SAND

There is only one type in this series. The series is confined to a range of sandy hills running diagonally from Pointe-a-Pierre in the south-west to Matura in the north-east, with few isolates elsewhere. Arena sand is found in the south-east of the present mapping unit.

Relief:

Generally hilly. It does sometimes occur in rolling country, but always in association with hills. If the valleys are permanently inundated, peaty sand occurs and this is separately mapped as Brazil peaty sand (*CHENERY* 1952). Brazil peaty sand does not occur in the Northern Range.

Parent Material:

Quaternary sands of Pleistocene age.

Profile Characteristics:

The morphology of Arena sand is simple. The topsoil is of variable brown sand — the darkness of the brown depending on the amount of humic matter in it. Under virgin forest the topsoil may be black. Like other siliceous soils in Trinidad, it may appear whitish during the dry season. In the ideal profile this colour gradually changes to orange brown and finally bright orange yellow within the first 5 feet. Under old forest, where the humic content is high and top horizon is black, the changes in colour profile may be masked in the top 4 feet. In deep

cuttings of the Arena sand-hills, stratification of the sand has been observed, and some of the layers are indurated. Iron compounds seem to be the cementing agent, and very small mica plates are common. The entire profile is sand.

Phase:

Deep.

Drainage:

Free to excessive.

Present Land Use:

Forest.

Nutrient Status:

Arena sand is extremely acid, and almost devoid of all major nutrients.

SOIL NO. 177: TALPARO SERIES

The Talparo series is very common in Central Trinidad. They are referred to as red clays. In fact it is the second horizon which is red. This layer has become exposed as a result of erosion and of the deep ploughing for sugar-cane cultivation over many years. They are the only red weathering clays in this study and are found near Matura.

Relief:

Hilly, intermediate uplands.

Parent Material:

Pliocene to upper Eocene clay and clay shale.

Profile Characteristics:

The topsoil is yellowish brown to dark brown. This horizon is followed by a variable grayish brown to yellowish brown layer — the dark colour possibly reflecting the amount of organic matter in the top few inches. Underlying this is a yellowish brown layer with a grayish cast and small red mottlings. The red mottlings increase with depth and in some profiles the red may become the main colour, being intermixed with greenish yellow or yellowish brown. Below this the greenish yellow or yellow brown clay persists, and brown and olive colours may appear with manganese dioxide stains, nests of gypsum and bits of clay shale.

The top horizon may be clay to silty clay but the lower horizons are always in the clay textural class. This soil is susceptible to land slides.

Phase:

Deep.

Drainage:

Impeded.

Present Land Use:

Mostly sugar cane, some of the other crops — citrus cocoa, vegetables and root crops — forest, especially teak.

Nutrient Status:

The Talparo series is moderately to strongly acid, well supplied with all the major nutrients. The presence of gypsum may meet some of the requirements for calcium, and it is reported that this soil fixes phosphate.

SOIL NO. 81: MARACAS SERIES

The Maracas series is one of two main soils found on the micaceous phyllites and schists of the Northern Range. The other series is the Matelot. These two series appear in several areas as an association. The outstanding and differentiating feature of the Maracas series is its red colour. There are two phases — a shallow and a deep. Observations indicate that crop responses on the Matelot and Maracas are very similar, and available analytical data have shown no appreciable difference. Nevertheless, the marked morphological difference in colour, possibly a function of iron content, supports the separation of the two series.

Relief:

Hilly, steep and gentle slopes.

Parent Material:

Phyllites, schists with some quartz.

Profile Characteristics:

Both deep (deeper than 48") and shallow (about 12") profiles are found. The topsoil is usually dark brown, under which a very marked reddish horizon exists. In many profiles the reddish horizon is particularly noticeable when the vegetative cover is thin and erosion has taken place. Light or deep red colours, mottled yellow and brown persist down into the parent material. The parent material which is weathering micaceous phyllites or schist varies in colour — red, black, silver and yellow. Pieces of phyllites or schists, which are commonly found in the profile, may have any of these colours and are easily identifiable on the red back-ground, along with sandstone or quartz fragments.

Phase:

Shallow, deep, stony.

Drainage:

Internal drainage free; run-off moderate to rapid.

Present Land Use:

Forest, cocoa, citrus, vegetables and root crops are successfully grown in the sheltered valleys. Slopes once cleared erode easily and become unproductive.

Nutrient Status:

This series is moderately low in all major nutrients although it is weakly leached, and is also moderately acid. A neutral member occurs where this soil is in close proximity to limestone deposits and then the pH value may approach neutrality.

SOIL NO. 181: MATELOT SERIES

The Matelot series includes both a deep and shallow profile, each of which may have a neutral member, and is developed from the micaceous phyllites or schists of the Northern Range. It is the yellow or yellowish brown soil extensively found in the uplands of Northern Trinidad.

In the shallow profile rock fragments of quartz, sandstone, phyllite or schist are found on the surface, particularly so, where erosion has been effective. The grass savannahs, commonly observed in the northern uplands of Trinidad,

are vegetative indications of the shallow Matelot or Maracas series. The Maracas series is the other extensive soil series of the Northern Range. It is believed that annual dry-season fires have contributed to the development of these savannahs.

The yellow soils of the Northern Range — Matelot series — and the reddish soils of the Northern Range — Maracas series — almost always occur as an association.

Relief:

Hilly, both steep and gentle slopes.

Parent Material:

Phyllites, schists with some quartz

Profile Characteristics:

The shallow profile may be 6 inches to bedrock. The topsoil is a dark yellow brown sandy loam underlain by a brown to yellow brown subsoil. Under natural forest the topsoil is black instead of brown, and the darker coloration persists throughout the profile. The parent material which is deeply weathered has a variety of colours — red, black, silver and yellow — which are inherent, and which may appear as mottlings, specklings or streaks. Small bits of phyllite, schist, quartz and sandstone are commonly found in the profile. Very often there is a predominant reddish yellow band just above the weathering rock.

The deep profile has similar topsoil coloration, but the subsoil is generally lighter in colour, and attains a depth, greater than 48 inches.

The texture of both profiles is normally sandy loam, but often becomes heavier in the lower horizons of the deep profiles.

Root penetration is very good: roots penetrate between the sheets of the weathering schist.

Phase:

Shallow, deep, stony.

Drainage:

Internal drainage free — encouraged by vertical and lateral movement along the phyllite and schist plates; run-off moderate to rapid.

Present Land Use:

Mostly forest; the valleys are planted in cocoa, citrus, with some vegetables and root crops.

Nutrient Status:

Moderately low in all nutrients except in the top horizon where the supply approaches adequacy. This may be due to the incorporation of nutrient-rich organic matter from leaf-litter back into the profile. The soil is moderately acid. When this soil develops in close proximity to limestone it may have a pH approaching neutrality.

SOIL NO. 281: SPRING HILL SERIES

The Spring Hill loams are a distinctive soil series of the Northern Range. The series normally occurs on micaceous talc schist deposits at elevations of about one thousand feet and in the heavy rainfall belt — above 100 inches mean annual precipitation. It is a fairly deep soil series, with an effective depth never less than 3 feet, and has a noticeable yellowish red colour with red mottlings. Two large areas of Spring Hill series are found — one at Spring Hill Estate on the Blanchisseuse Road and the other on the Anglais Road (Toco) near Moro Road. Cocoa is extensively grown on these soils especially on the moderate slopes.

Relief:

Hilly, moderately steep to very steep slopes.

Parent Material:

Micaceous talc schists.

Profile Characteristics:

A dark soil and a reddish brown to brownish yellow and olive subsoil, mottled red, make up the solum. The parent material is predominantly micaceous talc schist intermixed with soft sandstone and very little quartz. The deep solum normally 3 feet, and the absence of, or exceedingly low proportion of stones, uncommon to Northern Range soils, are two important characteristic features of the Spring Hill loams.

Phase:

Deep.

Drainage:

Internal drainage free.

Present Land Use:

Cocoa, coffee, citrus, bananas and forest.

Nutrient Status:

The entire profile is heavily leached and consequently is acid and low in all nutrients. It is a loam soil with good structure. The topsoil is slightly higher in nutrient level, and this can be due to nutrients held in the organic colloids. Organic matter is relatively high in the surface, with a C/N ratio of 12.0

SOIL NO. 82/L: DIEGO MARTIN SERIES

The Diego Martin series includes both a deep and a shallow phase. The soil is derived from calcareous schist of the Northern Range. The shallow phase usually occurs on the hill tops, while the deep phase is located on the slopes. This schist derived soil which is closely associated with the Maraval series has a dark topsoil.

Relief:

Hilly with steep slopes.

Parent Material:

Calcareous schists, phyllites, and rarely limestone.

Profile Characteristics:

The topsoil is very dark brown or grayish brown, with a good crumb structure which may also be present in the B horizon. The topsoil of the shallow

profile grades into olive colours, and finally into weathered laminated schist at about 18 inches. The schist is soft, soapy to the feel, and easy to crush. A feature of the shallow member of the series is the protrusion of schist above the surface of the soil.

The deep profile grades from dark brown into yellow and finally pale or light brown colours, while the parent material is encountered from 36 inches down.

Texture of the Diego Martin series varies between loam and sandy loam.

Phase:

Shallow, deep, stony.

Drainage:

Internal drainage free; run-off rapid. The topsoil in this series is not severely eroded.

Present Land Use:

These soils are used extensively and successfully for vegetable gardens. Emphasis is on chive, tomatoes and corn. Cocoa is grown in some areas. Most of the very steep areas are left in forest. Another use is quarrying.

Nutrient Status:

This soil is strongly alkaline throughout the profile. The percentage of free carbonate increases with depth. Nitrogen and phosphate are adequate but potassium is low. However, the nitrogen and phosphate appear to be carried on the organic complex. The soil is a loam in the upper horizons but becomes more sandy with depth. The increase in coarse grained material is partly due to the comminution of calcareous parent material and its incorporation into the profile.

SOIL NO. 182/L: LA PASTORA SERIES

The La Pastora series is developed from the massive type limestone of the Northern Range. It is distinct from the two other limestone derived soils in that it does not have the red colour of the Maraval nor the yellowish colour of the Platanal loam soils series. It is a dark soil with reddish colours in the subsoil, and is heavier in texture than the other limestone soils. There are two phases which follow the usual pattern of the Northern Range — shallow phase on the brows of the hills and deeper phase on the slopes. Despite the limestone parent material, some of the rock fragments found in the profile are non-calcareous. The series occurs only in the middle to high rainfall belt. Good exposures are found at La Pastora and at Aripo. The Aripo series described by WITT (1956) is evidently identical with the La Pastora loams.

They are productive soils on the milder slopes and are planted in cocoa; the steeper slopes are in forest.

Relief:

Hilly.

Parent Material:

Limestone.

Profile Characteristics:

The topsoil is dark brown, while the subsoil is reddish or yellowish brown. Rocks are not normally found in the solum but the C horizon always has rock fragments, some of which are non-calcareous although the soil is limestone derived.

The shallow phase found on the hilltops, and the deep phase on the slopes have similar profile characteristics.

Texture varies from light clay loam in the surface to a heavier clay loam in the lower horizon.

Root distribution is very good. Roots pass through the entire solum and are often found along the joint-faces of the parent limestone.

Phase:

Shallow, deep.

Drainage:

Free.

Present Land Use:

Cocoa, forest, some citrus on the gentle slopes. Quarrying is one of the important economic uses.

Nutrient Status:

The reaction is neutral in the surface, becoming moderately alkaline in the lower horizons. The soil is moderately well supplied with all major nutrients except potassium which is low.

SOIL NO. 282/L: MARAVAL SERIES

The Maraval series is one of the soils developed on limestone of the Northern Range. These soils occur on hilly land. On the slopes, the soil is usually deep but at the top of the hills the soil is shallow. In the shallow phase of the series the limestone may protrude above the surface of the soil. The deep and shallow soils are very closely associated and are therefore difficult to separate. The surface soil is always red or reddish brown although the red colour may sometimes be masked by organic matter. The Maraval series is widely distributed in the Saddle Road-Morne Coco Road districts where they are associated with another calcareous series — the Diego Martin soils.

Relief:

Hilly land, both steep and moderate slopes.

Parent Material:

Limestone, occasionally on calcareous schists.

Profile Characteristics:

The shallow phase has an effective depth to limestone of about 12 inches. The predominant colour is red. The topsoil is normally red to reddish brown although this colour may be masked by moisture and organic matter. In these cases the colour of the topsoil appears brown or gray. The characteristic red horizon then follows and is underlain by yellow brown or red brown layers which are part of the weathering material. In this weathering material the soil is not calcareous, but is intermixed with bits of limestone and schist.

In the deep phase, the reddish colour is persistent from top to bottom except for the darkening due to moisture and organic matter at the surface, and the presence of a narrow yellow or brown to olive horizon immediately above the limestone.

The texture of the profile is nearly uniform silt loam with a slight increase of clay content in the B horizon. In cases where the sand content of the parent limestone is high, the texture may become more sandy in the subsoil.

Phase:

Shallow, deep, stony.

Drainage:

Internal drainage free; run-off rapid.

Present Land Use:

Predominantly chive and vegetables. Cabbage and lettuce are grown during the wet season. All phases of the Maraval series are used for the same crops and there seems to be no difference in productivity. The upper slopes are used more intensively for agriculture than the lower slopes. Quarrying for the limestone is a common economic use.

Nutrient Status:

There is a nearly neutral reaction in the surface with an increase of alkalinity with depth. Phosphate is moderate at surface but very high in the lower horizons. All major nutrients are in adequate supply in the top soil except potassium which is low.

SOIL NO. 382/L: PLATANAL SERIES

This soil series is developed on limestone and is largely confined to the area in the medium to high rainfall belt in the south central part of the Northern Range — Platanal and Aripo.

The regions are hilly with moderately steep slopes. The soil developed on limestone in this area is dissimilar from previously described limestone soils. The differences are clearly defined in the profile characteristics. The deep soils on the moderate slopes are very suitable for cocoa and bananas, and in sections where the profile is shallow and limestone scattered at the surface, vegetative growth is still quite profuse even in the narrow non-stony areas.

Relief:

Hilly with some mild slopes.

Parent Material:

Limestone.

Profile Characteristics:

The topsoil is dark brown but where erosion has taken place it is yellowish brown. The subsoil is brown and yellowish brown with small quartz pebbles, and it breaks easily into subangular peds. The parent material is limestone.

The texture of the topsoil is loam and that of the subsoil is a silty loam to a clay loam. The entire profile has a well developed root-system except where

limestone inhibits penetration. Platanal soils are well drained and are among the most productive of the Northern Range soils.

Phase:

Shallow and deep.

Drainage:

Internal drainage free.

Present Land Use:

Cocoa, citrus, bananas, vegetables and forest.

Nutrient Status

The entire profile is alkaline except in the surface horizon where organic matter is high. Calcium, phosphate and magnesium increase with depth so that the soil should have a high supplying power for major nutrients. Nitrogen is good. Potassium is low.

SOIL NO. 91: SANS SOUCI SERIES

There should be no difficulty in recognising the Sans Souci series as it is the only group of soils developed on the igneous outcrops at San Souci in the north-eastern section of Trinidad. These soils are very productive and are nearly always under some form of agriculture — most commonly cocoa, vegetables and bananas.

Relief:

Hilly to rolling.

Parent Material:

Igneous rocks.

Profile Characteristics:

The surface soil is a dark grayish brown slightly sticky clay with a weak crumb structure and numerous roots. Below this layer the soil is dark yellow brown, and grades through a yellowish red horizon with brown mottlings to a brown horizon with yellowish red mottlings. In the solum soft rock fragments, coloured yellow brown or black are found—the yellow brown colour is the inherent colour of the igneous rocks, while the black colour is the result of organic coatings or staining. The soil has a distinctive blocky structure and breaks easily into angular peds. The texture gets heavier with depth. The parent material is weathered igneous rocks which are soft, easily crushed, predominantly dark yellow brown in colour with black stains. Sometimes calcareous skins occur along the crack-faces and fissures.

Phase:

Deep.

Drainage:

Internal drainage imperfect.

Present Land Use:

Cocoa, bananas, coffee, vegetables, ground provisions, tropical food and fruit trees.

Nutrient Status:

The soils are only slightly leached with a near neutral pH and an adequate supply of all the major nutrients, except potassium.

SOIL NO. 94/L: TOCO SERIES

The Toco series is derived from irregularly mixed calcareous and carbonaceous schist interbeds. The soil is obviously black in colour but its degree of blackness varies with the relative amount of black carbonaceous schist in the profile. Where the carbonaceous schist content is high, the inherent soil colour is much darker than where the carbonaceous schist content is low.

There is also a textural relationship in that the soil is heavier when the calcareous material is greater.

The series is considered as one of the most productive in Trinidad and is probably the highest yielder of *Musa* species in the territory.

Relief:

Hilly.

Parent Material:

Intermixture of calcareous and carbonaceous schists.

Profile Characteristics:

The members of this series which are developed predominantly on carbonaceous schist have a dark grayish brown to dark olive brown topsoil. This is followed by a horizon of olive brown to brownish yellow with dark brown and many blackish brown mottlings. The C horizon is black to dark gray mottled yellow brown.

As the carbonaceous schist content in the parent material decreases, the topsoil does not change but the black mottlings of the second horizon are not as marked, nor is the C horizon as black as the profile described in the preceding paragraph.

The texture is in the loam class and becomes heavier in the subsoil. The series is a group of very rich productive soils, and, only on extreme slopes are they not under economic crops. Even when rock outcrops are exposed at the surface, vegetative cover is still good.

Phase:

Moderately deep.

Drainage:

Internal drainage imperfect.

Present Land Use:

All commercial crops, forest on the steepest slopes.

Nutrient Status:

Toco soils are strongly alkaline in reaction, only very weakly leached with an adequate supply of all major nutrients except potassium. Phosphate values in the alkaline horizon are masked by the presence of free calcium carbonate but even in those horizons, where there is no free carbonate, the phosphate is still adequate. The sodium content is moderately high.

TABLE IV
SUMMARY OF SOIL DATA

SOIL	Altitude and Topography	Drainage Through Soil	Natural* Fertility	Factors Limiting Root Penetration	Erosion Hazard Under Arable crops	Land Use 1964-1965	Special Soil Management Problems
1. Cocal Sand	Flat Below 50'	Free	Low	None	None	Coconuts, secondary forest, recreational beaches	Desiccation
15. Caroni Peaty Clay	Flat Below 50'	Slow	High Saline	High water table	None	Swamp sedges, mangrove	Flooding for greater part of year, salinity
123. Guanapo Series	Flat Below 100' Trenches at higher elevations	Free	Low	Gravel in subsoil	None	Vegetables, citrus, cocoa	Droughtiness, stoniness
223. River Estate Series	Flat to gentle Below 100' Trenches at higher elevations	Free	Moderate	None	None	Housing, cocoa, coconuts, citrus, vegetables	Droughtiness
323. St. Joseph Series	Gentle Below 100' Trenches at higher elevations	Free	Moderate to low	Gravel in subsoil	Slight	Housing, citrus, cocoa	Droughtiness, stoniness
24/L. Grande Riviere Series	Flat Below 50'	Free	Moderate	None	None	Cocoa, bananas, vegetables, lastro	Occasional flooding, droughtiness
233. Sangre Grande Series	Flat Below 50' Trenches at higher elevations	Moderate	Moderate to high	None, peaty layer at 12'	None	Cocoa, citrus, bananas, plantains, vegetables	Flooding
39. Navet Clay	Flat Below 50'	Slow	Moderate to high	High water table	None	Swamp grass	Wet season flooding
41. Blanchisseuse Series	Moderate Below 1000'	Free	Moderate to low	Parent material at 36"	Moderate	Mainly coconuts, forest, some cocoa, tonca beans	Erodibility

SOIL	Altitude and Topography	Drainage Through Soil	Natural* Fertility	Factors Limiting Root Penetration	Erosion Hazard Under Arable Crops	Land Use 1964-1965	Special Soil Management Problems
141. Galera Sand	Gentle Below 500'	Free	Moderate to low	Gravel	Slight	Coconuts	Droughtiness, stoniness
241. Las Lomas Series	Gentle to moderate Up to 500'	Free	Low	None	Slight	Citrus, shifting cultivation	Droughtiness, occasional stoniness.
43. Acono Series	Gentle to moderate and flat ledges Up to 500'	Free	Low	Gravelly and bouldery subsoil	Slight to moderate	Cocoa, citrus, tonca-beans, shifting cultivation	Droughtiness, stoniness
143. St. Augustine Series	Gentle to moderate Below 100'	Free	Low	Gravelly and bouldery subsoil	Slight	Housing, citrus, some sugar cane, vegetables, shifting cultivation	Droughtiness, stoniness
243. Santa Cruz Series	Gentle to moderate Up to 500'	Free	Moderate to low	Gravelly and bouldery subsoil	Moderate	Citrus, cocoa	Slight droughtiness, stoniness
53. Anglais Series	Moderate to steep enclosed or partially enclosed areas Up to 1500'	Moderate	Moderate to low	None	Slight to moderate	Cocoa, bananas, plantains, coffee, tonca beans,	Erodibility
153. Cleaver Series	Gentle to moderate Up to 250'	Slow	Very low	Stony and bouldery subsoil	Slight	Scrub, forest	Droughtiness, drainage
55. Piarco Series	Flat to gently undulating Below 250'	Free in topsoil, slow in subsoil	Very low	Pan Gleisation	Slight	Scrub, forest	Droughtiness, drainage

SOIL	Altitude and Topography	Drainage Through Soil	Natural* Fertility	Factors Limiting Root Penetration	Erosion Hazard Under Arable Crops	Land Use 1964-1965	Special Soil Management Problems
255. Valencia Series	Flat to gently undulating Below 250'	Free in topsoil, very slow in subsoil	Very low	Pan Gleisation	Slight	Scrub, forest	Droughtiness, drainage
59. Long Stretch Series	Flat to gently undulating Below 250'	Free in topsoil, very slow in subsoil	Very low	Pan Gleisation	Slight	Scrub, forest	Droughtiness, drainage
261. Arena Sand	Hilly Up to 500'	Free	Low	None	Moderate	Scrub, forest, tobacco	Droughtiness
177. Talparo Series	Hilly Up to 500'	Slow	Moderate to high	Heavy clay in top 48"	Moderate to severe landslips	Sugar, teak forest, virgin forest	Desiccation in dry season
81. Maracas Series	Moderate to very steep 100'-3000'	Free	Moderate to low	None, parent material in shallow phase	Moderate to very severe	Cocoa, coffee, citrus, bananas, vegetables, tonca beans, forest	Erodibility, occasional stoniness
181. Matelot Series	Moderate to very steep. 100'-3000'	Free	Moderate to low	None, parent material in shallow phase	Moderate to very severe	Cocoa, coffee, citrus, bananas, vegetables, tonca beans, forest.	Erodibility, occasional stoniness
281. Spring Hill Series	Moderate to very steep 100'-2000'	Free	Moderate to low	None	Severe	Cocoa, coffee, citrus, bananas, forest.	Erodibility, occasional stoniness

SOIL	Altitude and Topography	Drainage Through Soil	Natural* Fertility	Factors Limiting Root Penetration	Erosion Hazard Under Arable Crops	Land Use 1964-1965	Special Soil Management Problems
82/L. Diego Martin Series	Moderate to very steep 100'-2000'	Free	Moderate	None, stoniness in shallow phase	Moderate to very severe	Green vegetables, chive, cocoa, forest, scrub	Erodibility stoniness
182/L. La Pastora Series	Moderate to very steep 100'-2000'	Free	Moderate to low	None	Severe	Cocoa, citrus, forest.	Erodibility, occasional stoniness
282/L. Maraval Series	Moderate to very steep 100'-2000'	Free	Moderate to high	None, stoniness in shallow phase	Moderate to very severe	Green vegetables, chive, cocoa, forest, scrub	Erodibility, occasional stoniness
382/L. Platanal Series	Moderate to very steep 500'-3000'	Free	Moderate to high	None	Moderate to very severe	All normal crops, forest	Erodibility, occasional stoniness
91. Sans Souci Series	Moderate to very steep 100'-2000'	Moderate	Moderate to high	Partly weathered rock in subsoil	Moderate to very severe	All normal crops, forest	Erodibility
94/L. Toco Series	Hilly to very steep 100'-2000'	Moderate	Moderate to high	Partly weathered rock in subsoil	Moderate to very severe	All normal crops, forest	Erodibility, occasional stoniness

*Levels based on soil analytical data compared with standards in Appendix I.

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APPENDICES

APPENDIX I

Sampling Procedures and Analytical Methods

Pits were dug in each soil series and soil samples taken from each horizon.

Texture of schistose soils, assessed in the field by "finger and thumb" was more coarse grained than indicated by laboratory measurement. Probably some of the coarser schist grains broke down into much smaller particles when they were ground to pass a 2mm. sieve, prior to analysis

Analyses were done using the standard methods of the Central Analytical Laboratory of the Regional Research Centre,⁽¹⁾ University of the West Indies and the Soils Division⁽²⁾ of the Central Experiment Station of the Ministry of Agriculture, Trinidad and Tobago.

Exchangeable potassium in all the schistose soils was low. Although no information was available about rate of release of potassium, apparently it was quick enough to maintain adequate levels of available potassium. Unpublished data by Ahmad⁽³⁾ certainly showed that levels of total potassium in these soils were satisfactory, while unpublished data by Brown *et.al.*⁽⁴⁾ indicated that factors such as wetting and drying probably have some effect on availability.

In some of the analytical data tables which follow, those measurements marked with an asterisk show that total exchangeable bases (T.E.B.) are sometimes greater than cation exchange capacity (C.E.C.); reasons for this are not yet known. Perhaps either C.E.C. values in some soils were under-estimates or cations, especially Ca^{++} and Mg^{++} , other than exchangeable ones were brought into solution when T.E.B. determinations were made.

(1) Lang, D. M. *et al* (1963) — see references

(2) Chenery, E. M. (1952) — see references

(3) Ahmad, N. (1946) *University of the West Indies, Trinidad*

Analysis of schistose soils of the Northern Range, Trinidad showed:—

2.08 to 3.33% K in silt fraction

2.50 to 3.33% K in clay fraction

(4) Brown, C. B., Clemanty, J., Oliver, F. and Forde, K (1965)

Central Experiment Station, Ministry of Agriculture, Trinidad and Tobago.

LEVELS USED FOR DETERMINING NUTRIENT STATUS⁽¹⁾
USUAL UNITS

Description	C	N	C/N	C.E.C.		T.E.B.	Ca	Mg	K	Na	B.S.	Acid Sol. P ⁽²⁾
				mineral soils	peats							
High	20	1.0	25	40	80	25	20	8	1.2	2.0	81-100	35
	10-20	0.5-1.0	15-25	25-40	50-80	15-25	10-20	3-8	0.6-1.2	0.7-2.0	61-80	22-35
Moderate	4-10	0.2-0.5	10-15	12-25	30-50	7-15	5-10	1.0-3	0.3-0.6	0.3-0.7	41-60	15-22
	2-4	0.1-0.2	8-10	6-12	20-30	3-7	2-5	0.3-1.0	0.2-0.3	0.1-0.3	20-40	7-15
Low	2	0.1	8	6	20	3	2	0.3	0.2	0.1	20	7

LEACHING

Average base Saturation of profile	Description
70-100	Very weakly leached
50-70	Weakly leached
30-50	Moderately leached
10-30	Strongly leached
10	Very strongly leached

pH

Value	Description
8.0	Strongly alkaline
7.6-8.0	Moderately alkaline
7.1-7.5	Slightly alkaline
6.6-7.0	Near neutral
6.0-6.5	Slightly acid
5.3-5.9	Moderately acid
4.5-5.2	Strongly acid
4.5	Extremely acid

(1) DESCRIPTION FOR ANALYTICAL FIGURES USED BY N.Z. SOIL SURVEY
AND FIJI SOIL SURVEY, MARKED (2)

TABLE V
FIELD DESCRIPTION AND ANALYTICAL DATA OF

<i>Soil</i>	<i>Page</i>
1.: Cocal Sand	60
15.: Caroni Peaty Clay	61
123.: Guanapo Series	62
223.: River Estate Series	63
323.: St. Joseph Series	64
24/L.: Grande Riviere Series	65
233.: Sangre Grande Series	66
39.: Navet Clay	67
41.: Blanchisseuse Series	68
141.: Galera Sand	69
241.: Las Lomas Series	70
43.: Acono Series	71
143.: St. Augustine Series	72
243.: Santa Cruz Series	73
53.: Anglais Series	74
153.: Cleaver Series	75
55.: Piarco Series	76
255.: Valencia Series	77
59.: Long Stretch Series	78
261.: Arena Sand	79
177.: Talparo Series	80
81.: Maracas Series	81
181.: Matelot Series	82
281.: Spring Hill Series	83
82/L.: Diego Martin Series	84
182/L.: La Pastora Series	85
282/L.: Maraval Series	86
382/L.: Platanal Series	87
91.: Sana Souci Series	88
94/L.: Toco Series	89

SOIL UNIT: 1

COCAL SAND

LOCATION: Scotland Bay, flat 80 yds from shore line
DRAINAGE: Free to excessive internal drainage
VEGETATION: Secondary forest

RELIEF: Flat
SLOPE: A
EROSION: O

* Botanical names of most of the plants under Vegetation can be found in Useful and Ornamental Plants of Trinidad and Tobago, R. O. Williams (1951) Guardian Printers, Trinidad, W.I.

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Very dark brown 10YR 2/2 loamy sand — loose — weak medium and fine crumbs — numerous roots — abrupt smooth boundary to:
3 — 9 Grey 10YR 5/1 and dark grayish brown 10YR 4/4 loamy sand — few roots — loose — some humic material as coating on sand — clear smooth boundary to:
9 — 17 Pale olive 5Y 6/3 and brown 7.5YR 5/2 sandy loam — few roots — slightly compacted in site — loose — easy to crush when removed — abrupt smooth boundary to:
17 — 30 Light gray 7.5YRN 7/ and dark brown 7.5YR 3/2 loamy sand — few roots — slightly compacted in site — loose — easy to crumble when removed — abrupt smooth boundary to:
30 — 38 Light gray 10YR 7/1 and strong brown 7.5YR 5/6 loamy sand — compact in place — easy to crush when taken out of pit — very few roots — abrupt wavy boundary to:
38 White 5Y 8/2 and light brown 7.5YR 6/4 gravelly sandy loam — no roots — difficult to dig, but easy to crush when taken out of pit — 50% gravel, quartz and some sandstones.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 - 3	4.0		194	11	70	6	13	5.7	1.8	1.0	0.8	0.12	0.14	32	1.6	0.14	11	8
3 - 9	4.4		46	11	75	5	14	2.0	0.2	0.2	0.1	0.02	0.00	10	1.0	0.05	20	2
9 - 17	5.0		20	10	64	4	15	1.9	0.2	0.2	0.1	0.01	0.00	11	0.8	0.03	27	1
17 - 30	5.2		15	11	74	7	12	1.3	0.4	0.4	0.1	0.02	0.10	31	0.4	0.03	13	Trace
30 - 38	5.2		12	9	74	7	15	1.3	0.4	0.2	0.3	0.01	0.00	31				1
38 +	5.2		16	9	65	5	20	1.7	0.4	0.2	0.3	0.00	0.17	24				2

SOIL UNIT: 15

CARONI PEATY CLAY

LOCATION: Swamp
DRAINAGE: Perennial inundation — impeded
VEGETATION: Mangrove

TOPOGRAPHY: Flat
SLOPE: A
EROSION: O to +

PROFILE DESCRIPTION

Depth (ins.)

0 — 6 Very dark gray 7.5 YR N3/ clay — numerous roots interlaced with each other — very sticky:
6 — 12 Very dark gray 2.5 Y N3/ clay — sticky — many thin mangrove roots:
12 — 18 Very dark gray 10 YR 3/1 clay — sticky — many thin roots:
18 — 30 Black 10 YR 2/1 peaty clay — sticky — many thin roots:
30 — 42 Very dark gray 7.5 YR N3/ clay — numerous roots interlaced with each other — very sticky.
42 — 60 Black 7.5 R N2/ and very dark brown 10 YR 2/2 peaty clay — many thin roots — definite increase in peaty material:

These samples were taken with a grain sampler so that the boundaries could not be clearly defined.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 - 6	4.8			4	2	18	48								15.9	.59	27	SAL- INE
6 - 12	5.1			7	6	16	43								16.2	.56	29	
12 - 18	5.0			4	13	16	40								15.8	.24	66	
18 - 30	4.5		S A L I N E	—	—	—	—	S	A	L	I	N	E		24.2	.65	37	
30 - 42	4.2			1	3	16	40								—	—	—	
42 - 60	3.8			2	7	7	33								—	—	—	

SOIL UNIT: 123

GUANAPO SERIES

LOCATION: De Matas Estate — Acono Road
DRAINAGE: Free
VEGETATION: Cocoa, ferns and cockspur

TOPOGRAPHY: Flat
SLOPE: A
EROSION: O

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Very dark gray brown 10 YR 4/2 sandy loam — weak medium crumbs — slightly sticky — numerous roots — micaceous — friable — smooth clear boundary to:

4 — 14 Grayish brown 10 YR 5/2 — silt loam — numerous thin roots — smooth loamy feel due to micaceous particles — some clay — slightly sticky — structure of weak medium sub-angular peds — smooth clear boundary to:

14 — 26 Dark yellow brown 10 YR 4/4 sandy clay loam — breaking into coarse and medium sub-angular blocky peds — few roots — very micaceous — friable — slightly sticky — abrupt wavy boundary to:

26 — 40 Brown 7.5 YR 5/2 gravelly sand — very micaceous — few roots — small rounded pebbles comprising of quartz and sandstone — some shiny schist particles — gravel make up about 50% of horizon — smooth gradual boundary to:

40 Strong brown 7.5 YR 5/6 gravelly sandy loam — small roots going through the gravel — some large quartz and sandstone rocks — dark and silvery schist particles — gravel make up about 80% of horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	*C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 - 4	5.0		261	2	52	22	25	6.3	9.5	7.9	1.2	0.08	0.00	100	1.8	0.22	8.2	10
4 - 14	4.8		55	2	49	24	27	5.2	3.2	2.4	0.6	0.02	0.02	62	0.6	0.08	7.5	3
14 - 26	5.1		43	7	49	16	29	5.1	3.1	2.6	0.6	0.01	0.30	61	0.6	0.08	7.5	2
26 - 40	5.3		26	44	39	6	15	2.5	1.5	1.0	0.3	0.00	0.23	60				8
40+	5.4		38	42	37	12	13	3.3	2.1	2.0	0.9	0.00	0.13	64				9

RIVER ESTATE SERIES

TOPOGRAPHY: Flat
SLOPE: A
EROSION: O

63

0 — 3	Dark olive gray 5 Y 3/2 sandy loam — weak medium crumbs — numerous roots at the top interlaced with each other — loose — micaceous — smooth feel — with some dark brown stains due to organic matter — smooth clear boundary to:
3 — 9	Yellow brown 10YR 5/4 sandy loam — micaceous — breaking into weak angular and sub-angular blocky peds — some small flat dark particles — small amount of fine fragments — numerous roots — clear wavy boundary to:
9 — 21	Dark yellow brown 10 YR 3/4 and olive gray 5 Y 4/2 gravelly loam — small flat sandstone — many roots — the dark colour may be due to organic matter — smooth clear boundary to:
21 — 32	Brown 7.5 YR 3/2 and grayish brown 2.5 Y 5/2 silty loam — micaceous — many roots — soil very smooth to the feel — no gravel — gradual wavy boundary to;
32 — 50	Light olive gray 5Y 6/2 and dark yellowish brown 10YR 3/4 loam — micaceous — few roots — dark streaks of manganese dioxide — no gravel — smooth clear boundary to:
50	Pale olive 5Y 6/4 and dark brown 10YR 3/3 sandy loam — micaceous — smooth to the feel — very moist — few roots — absence of gravel — coarse sand grains.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m P
				Crse. Sand	Fine Sand	Silt	Clay	*C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 - 3	6.6		226	12	54	16	18	9.3	10.3	10.3	1.1	0.05	0.05	100	2.0	0.26	7.7	114
3 - 9	6.6		52	20	56	9	16	3.2	3.8	3.4	0.4	0.02	0.01	100	0.7	0.08	8.0	35
9 - 21	6.6		37	19	58	8	15	2.4	2.2	1.8	0.3	0.01	0.01	92	0.7	0.07	10.0	15
21 - 33	6.3		35	2	49	18	34	5.8	4.8	4.7	0.3	0.03	0.01	87	1.1	0.13	8.5	6
33 - 50	6.3		30	1	48	18	34	4.6	3.9	4.0	0.3	0.02	0.09	85				5
50+	6.5		26	3	55	18	30		2.7	2.9	0.3	0.00	0.01	75				6
								Phosphate recently added										

SOIL UNIT: 323
LOCATION Peasant Garden — Curepe — Bank of the St. Joseph River
DRAINAGE: Free to excessive
VEGETATION: Vegetables, wild coffee, broom grass, savannah grass,
 Spanish needle.

ST. JOSEPH SERIES
TOPOGRAPHY: Flat
SLOPE: A
EROSION: O

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Dark yellow brown 10YR 3/4 sandy loam — numerous fine and medium roots — weak medium and coarse crumbs — loose — some burnt material — abrupt smooth boundary to:
 4 — 20 Yellowish brown 10YR 5/6 and very dark brown 10YR 2/2 sandy loam — numerous roots — structure of weak fine sub-angular peds — soil very loose — bits of micaceous particles — the dark colour is due to organic matter leached down from above — smooth clear boundary to:
 20 — 30 Olive brown 2.5Y 4/4 loamy sand — micaceous — loose — many roots — some white shiny particles — smooth clear boundary to:
 30 — 40 Brown 10YR 5/3 loamy sand — loose — white shiny particles — thin sandier layer with many dark flat particles — few fine roots — clear wavy boundary to:
 40 — 60 Olive 5Y 4/4 and brown 10YR 3/3 loamy sand — loose — some black soft specks, probably carbon — some live roots — old root traces — smooth clear boundary to:
 60 Yellow brown 10YR 5/4 gravelly loam — few roots — micaceous — very small black and white, flat particles — gravel of rounded sandstone and quartz pebbles, which make up 80% of the horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	*C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	6.0		89	7	68	12	13	2.9	3.4	4.0	0.5	0.01	0.01	100	0.9	0.11	8.2	32
4 — 20	6.6		40	4	58	18	21	4.4	4.4	4.0	3.0	0.01	0.04	100	0.7	0.08	8.8	33
20 — 30	6.6		38	6	73	8	13	3.2	3.0	3.0	0.6	0.00	0.01	94	0.6	0.08	7.5	34
30 — 40	7.0		34	13	67	8	9	1.7	1.4	3.0	0.5	0.00	0.00	82				12
40 — 60	6.8		76	4	71	9	15	2.7	2.0	3.0	0.6	0.00	0.02	74				29
60 +	6.7		54	31	44	14	9	2.7	2.4	2.0	1.0	0.00	0.00	100				28

SOIL UNIT: 24/L

GRAND RIVIERE SERIES

LOCATION: Grand Riviere Village — West of the River
 DRAINAGE: Free to excessive
 VEGETATION: Abandoned cocoa, ferns, corn, bananas and various kinds of shrubs and grasses.

TOPOGRAPHY: Flat
 SLOPE: A
 EROSION: O

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Dark gray 5Y 4/1 sandy loam — bits of shiny white particles (probably mica) — smooth feel — weak angular crumbs — slightly sticky — schist fragments — micaceous — many roots — smooth abrupt boundary to:
 4 — 18 Olive gray 5Y 4/2 micaceous sandy loam — numerous thin roots — structure of weak angular crumbs — humic matter in root traces — some dead root traces — few worm casts — smooth clear boundary to:
 18 — 38 Gray 10YR 6/1 mottled fine faint light olive gray 5Y 6/2 sandy loam — breaking into weak crumbs — very micaceous — loose — some darker flute like areas that have filtered from the top — smooth abrupt boundary to:
 38 — 52 Gray 5Y 6/1 micaceous sandy loam — smooth feel due to micaceous material — many small roots — many flat black 5Y 2/1 schist particles — loose — soft yellow 10YR 7/6 sandstone particles — many small shiny particles — some quartz pebbles — smooth abrupt boundary to:
 52 Light olive gray 5Y 6/2 and pale yellow 5Y 7/3 micaceous loam — loose — smooth feel — many rock particles, consisting of quartz and sandstone — some flat slate-like soft fragments which make up about 40% of horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		125	5	53	22	24	9.0	6.5	4.8	1.9	0.39	0.51	72	0.9	0.15	6.0	34
4 — 18	4.9		138	2	47	21	29	8.2	4.0	3.5	0.7	0.10	0.73	49	1.4	0.12	11.7	31
18 — 38	4.9		70	11	45	18	32	10.5	2.3	2.1	0.9	0.06	0.52	22				6
38 — 52	4.9		124	18	46	12	24	4.7	3.0	2.0	0.9	0.05	0.20	64				8
52 +	5.4		41	34	32	14	24	3.5	3.2	2.8	0.6	0.05	0.28	91				12

SOIL UNIT: 233

SANGRE GRANDE SERIES

LOCATION: El Repose Demonstration Station
 DRAINAGE: Imperfect, intermittent floods
 VEGETATION: Cocoa, bananas, cassava, razor grass and savannah grass

TOPOGRAPHY: Flat
 SLOPE: A
 EROSION: O to t

PROFILE DESCRIPTION

Depth (ins.)

0 — 3

Dark brown 7.5YR 4/4 mottled few fine distinct strong brown 7.5YR 5/6 and few medium distinct olive gray 5Y 4/2 clay — silty to touch — numerous roots — weak coarse crumbs — few fine tubular pores — slightly sticky — slightly plastic — abrupt smooth boundary to:

3 — 12

Gray 5Y 6/1 and grayish brown 2.5Y 5/2 mottled few fine distinct yellowish brown 10YR 5/6 silty clay — many roots — few root traces — few fine tubular pores — weak irregular angular blocky peds — few dark gray 10YR 4/1 infilled tunnels — slightly sticky — slightly plastic — firm — smooth clear boundary to:

12 — 24

Light gray 7.5YR N7/ mottled few medium distinct strong brown 7.5YR 5/4 and few fine faint greenish yellow 5GY 7/1 clay — many roots — few root traces — dead roots — many fine tubular pores — few manganese dioxide stains — slightly sticky — slightly plastic — mottles are usually sandy — smooth clear boundary to:

24 — 40

Bluish gray 5BG 6/1 mottled few medium prominent dark brown 7.5YR 5/4 clay — few fine roots — few root traces — few dead roots — weak irregular angular peds — few tubular pores — mottles are sandy — few fine manganese dioxide stains — sticky — plastic — compact — firm — gradual smooth boundary to:

40 — 58

Bluish gray 5BG 7/1 mottled few medium distinct reddish yellow 7.5YR 6/6 and few coarse distinct brown 7.5YR 5/4 clay — few fine roots — few dead roots — few fine pores — weak irregular angular blocky peds — slightly sticky — plastic — mottles are sandy — compact — firm — irregular gradual boundary to:

58

Bluish gray 5BG 6/1 mottled few coarse prominent reddish brown 5YR 4/3 and few fine faint yellowish brown 10YR 4/4 silty clay loam — few fine roots — few dead roots — few fine pores — weak irregular angular blocky peds — slightly sticky — plastic — compact — firm — mottles are sandy.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.8		119	0	32	25	41	11.8	5.9	3.7	1.5	0.06	0.07	50	2.4	0.28	8.6	13
3 — 12	4.7		47	0	39	16	42	10.7	2.5	1.4	0.7	0.01	0.11	23	0.9	0.12	7.5	3
12 — 24	4.9		37	0	31	18	42	11.5	2.1	1.3	0.7	0.02	0.11	18				1
24 — 40	5.0		37	0	42	14	41	10.5	2.6	1.3	1.2	0.02	0.13	25				8
40 — 58	5.3		25	0	45	16	38	10.6	2.9	0.6	1.6	0.02	0.11	27				3
58	5.4		21	1	54	12	34	7.3	2.5	0.5	1.1	0.01	0.12	34				3

SOIL UNIT: 39

NAVET CLAY

LOCATION: Fishing Pond Road

DRAINAGE: Impeded.

VEGETATION

Recently cleared secondary forest, citrus, cassava, ochro, (okra)
corn grass, razor grass, congo-lala, broom grass, wild mustard,
porters bush, and ferns.

TOPOGRAPHY: Flat

SLOPE: A

EROSION: O to +

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark brown 10YR 3/2 clay — numerous fine roots — worms and worm casts — weak medium and coarse crumbs — slightly sticky — slightly plastic — abrupt smooth boundary to:

2 — 8

Gray 10YR 5/1 and grayish brown 10YR 4/2 clay — many fine roots — old root channels — structure of weak medium and coarse sub-angular peds — slightly sticky — slightly plastic — smooth clear boundary to:

8 — 18

Gray 5Y 5/1 mottled few medium distinct reddish yellow 7.5YR 6/8 clay — few fine roots — few dead roots — few root channels — few fine tubular pores — structure of weak medium sub-angular peds — sticky — plastic — compact — firm — ped faces stained grayish brown 10YR 5/2 — smooth clear boundary to:

18 — 36

Light gray 2.5Y N7/ mottled few medium distinct yellowish brown 10YR 5/8 clay — few medium and fine roots — dark yellow brown 10YR 4/4 root channels — structure of weak coarse sub-angular peds — sticky — plastic — compact — firm — gradual smooth boundary to:

36 — 56

Greenish gray 5G 6/1 mottled few fine distinct strong brown 7.5YR 5/6, few coarse distinct red 10Y 5/6 clay — few fine roots — dead roots — root channels — sticky — plastic — compact — firm — smooth boundary to:

56

Bluish gray 5BG 7/1 mottled few coarse distinct yellowish brown 10YR 5/6, few medium distinct red 7.5 R 4/8 clay — very few fine roots — few dead roots — sticky — plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	5.3		127	2	10	29	56	31.6	19.3	12.7	7.0	0.88	1.53	61	8.1	0.57	14.2	16
2 — 8	5.3		54	0	8	29	66	19.7	11.2	4.8	7.3	0.28	1.57	57	2.9	0.29	10.0	10
8 — 18	4.8		84	0	6	26	68	17.7	8.4	3.3	6.3	0.14	1.50	47	0.9	0.14	6.4	5
18 — 36	4.7		475	0	5	26	72	16.7	10.9	4.1	9.2	0.12	2.92*	55				6
36 — 56	5.9		749	0	3	24	74	18.7	16.2	5.6	12.3	0.12	5.95*	87				3
56 +	6.8		636	0	5	21	76	17.5	16.2	5.3	11.4	0.08	6.29*	93				4

*Probably includes water soluble sodium

SOIL UNIT: 41

LOCATION: Paria Road west of Blanchisseuse
 DRAINAGE: Free
 VEGETATION: Coconuts interplanted with limes, para grass, vinvin and some hard wood shrubs

BLANCHISSEUSE SERIES

TOPOGRAPHY: Low undulating hills
 SLOPE: C
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Very dark brown 10YR 2/2 sandy loam — many coconut roots, interlaced with each other — soil breaking into weak medium crumbs — easy to crush — slightly sticky — smooth clear boundary to:
 4 — 9 Light brownish gray 10YR 6/2 and dark brown 10YR 3/3 mottled few fine distinct red 2.5YR 4/8 clay loam — breaking into weak irregular sub-angular peds — humic matter filtered down root traces — many fine and medium roots — old root traces — slightly sticky — slightly plastic — smooth clear boundary to:
 9 — 21 Grayish brown 2.5Y 5/2 mottled common medium distinct red 10R 4/6 and few fine faint brownish yellow 10YR 6/6 clay — many fine roots — structure of irregular sub-angular peds — slightly sticky — slightly plastic — small schist and sandstone fragments — abrupt wavy boundary to:
 21 — 31 Gray 2.5YR N5/, dusky red 10R 3/3 and reddish yellow 5YR 6/6 mottled faint fine white 7.5YR 8/2 clay loam — some schist and sandstone — schist incorporated in the soil — few thin roots — some quartz pebbles — smooth clear boundary to:
 31 — 38 Gray 2.5YR N5/, very dark gray 5YR 4/1 and red 7.5R 5/6 clay loam — few roots — schist fragments and bits of sandstone and quartz — smooth clear boundary to:
 38 Very dusky red 7.5R 2/2 and gray 2.5YR N5/ mottled few fine faint brownish yellow 10YR 6/8 schist with some soil — one large coconut root — more schist than soil — some quartz pebbles — schist material and rocks make up about 55% of horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.2		192	21	27	17	40	14.8	14.3	8.4	6.2	0.49	1.18	97	2.2	0.41	5.4	32
4 — 9	5.2		62	16	26	13	51	11.2	8.4	4.6	3.9	0.13	1.00	75	1.9	0.21	9.0	8
9 — 21	4.9		62	6	19	17	62	13.2	7.1	3.8	3.6	0.11	1.52	54	0.8	0.17	4.7	3
21 — 31	5.1		80	5	17	23	59	13.7	8.8	5.0	3.6	0.09	2.62	64				2
31 — 38	5.3		94	8	18	25	52	12.8	9.1	4.9	3.6	0.10	2.88	71				2
38	5.4		93	4	18	25	50	12.2	8.0	4.0	3.5	0.08	3.08	66				7

SOIL UNIT: 141

GALERA SERIES

LOCATION: Large coconut estate — opposite Balandra Bay Road TOPOGRAPHY: Slightly undulating.
 DRAINAGE: Free with water seepage SLOPE: D
 VEGETATION: Coconuts, black sage, razor grass, caraquite and other grasses EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 4

Dark brown 7.5YR 3/2 sandy loam with a small amount of clay material — weak angular crumbs — many roots thick, thin and live — some shiny quartz grains — one large quartz rock — slightly sticky — abrupt smooth boundary to:

10 — 23

Dark grayish brown 10YR 4/2 coarse sandy loam — slightly sticky — breaking into weak medium and coarse sub-angular blocky peds — many coconut roots — some small quartz grains — smooth clear boundary to:

23 — 31

Dark gray to grayish brown 10YR 4/1 coarse sandy loam — sand grains becoming larger — few roots — moist — some clay, which binds the sand together — structure of angular peds — smooth clear boundary to:

31 — 46

Dark grayish brown and yellow brown 10YR 4/4 to 5/6 gravelly loam — small rounded quartz gravel — water seeping into pit — few roots — smooth clear boundary to:

46

Dark yellow brown 10YR 4/4 mottled yellow brown 5/8 coarse sand to gravelly loam — compact — water seeping into pit — few roots — smooth clear boundary to:

Yellow brown 10YR 5/8 and light gray 10YR 6/1 mottled light red 2.5YR 6/6, gravelly loam — water seeping into pit from the sides.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.0		228	61	14	12	14	4.3	3.1	3.0	2.0	0.06	0.28	72	3.5	0.27	13.0	4
4 — 10	5.0		129	62	12	10	16	2.1	1.4	2.0	0.7	0.10	0.19	67	1.8	0.15	12.0	2
10 — 23	5.0		40	55	15	14	16	2.5	1.2	2.0	0.4	0.01	0.14	48	1.2	0.11	10.9	2
23 — 31	5.0		26	57	13	12	11	2.7	1.0	2.0	0.4	0.01	0.22	37				1
31 — 46	5.0		21	54	15	10	20	3.0	0.6	2.0	0.2	0.01	0.08	20				1
46 +	5.0		22	54	14	4	28	6.6	0.4	1.0	0.1	0.01	0.08	6				2

SOIL UNIT: 241

LAS LOMAS SERIES

LOCATION: North west of Centeno Experiment Station Buildings
 DRAINAGE: Free
 VEGETATION: Savannah grass, water grass, Christmas bush, black sage.
 para grass, broom grass

TOPOGRAPHY: Gently rolling
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Very dark grayish brown 10YR 3/2 sandy loam — numerous fine and medium roots — weak medium and coarse crumbs — few root channels — fine pores — bits of burnt material — smooth clear boundary to:
 4 — 9 Dark yellowish brown 10YR 4/4 and very dark gray 10YR 3/1 sandy loam — many fine and medium roots — weak medium and coarse crumbs — few root channels — loose — smooth clear boundary to:
 9 — 21 Yellowish brown 10YR 5/6 and reddish yellow 5YR 6/8 loam — many fine roots — root channels — few fine tubular pores — crack faces stained grayish brown 10YR 5/2 — loose — gradual smooth boundary to:
 21 — 49 Red 2.5YR 5/6 mottled few medium faint strong brown 7.5YR 5/6 sandy loam — many fine roots — many fine medium tubular pores — few root channels — infilled tunnels brown 10YR 5/3 — structure of weak fine angular peds — abrupt smooth boundary to:
 49 — 61 Strong brown 7.5YR 5/6 and yellowish brown 10YR 5/8 sandy loam — few fine and medium roots — few dead roots — few root channels — few fine tubular pores — structure of weak and medium angular blocky peds — loose — gradual smooth boundary to:
 61 Very pale brown 10YR 7/3 and reddish yellow 7.5YR 6/6 mottled few fine faint red 2.5YR 4/6 sandy loam — very few fine roots — few dead roots — few fine tubular pores — stains of grayish brown 10YR 5/2 — loose.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.3		58	10	64	11	17	4.6	2.0	1.9	0.4	0.04	0.16	43	1.6	0.12	13.3	6
4 — 9	4.9		50	8	65	9	21	4.1	2.0	1.8	0.5	0.03	0.22	49	1.5	0.11	13.6	4
9 — 21	4.5		43	8	64	7	25	4.7	0.6	0.5	0.2	0.03	0.23	13	0.6	0.08	7.5	4
21 — 49	4.5		28	5	60	5	31	4.7	0.4	0.2	0.2	0.03	0.09	9				4
49 — 61	4.7		25	7	69	7	21	3.0	0.4	0.3	0.1	0.03	0.15	13				0
61	4.6		28	4	76	7	19	2.5	0.2	0.0	0.1	0.02	0.03	8				0

LOCATION SOIL UNIT: 43
Acono Road South of De Matas Estate

DRAINAGE: Free

VEGETATION: — Cocoa, pois doux, orange, tonca beans, savannah grass and black sage.

ACONO SERIES

TOPOGRAPHY: Mildly undulating foothills of the Northern Range.

SLOPE: C

EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 4

Very dary brown 10YR 2/2 sandy loam — weak fine and medium crumbs — slightly sticky — numerous roots some shine grains — probably sand grains — bits of quartz as small pebbles — smooth clear boundary to:

4 — 12

Dark brown 10YR 3/3 sandy clay loam — weak, medium and coarse sub-angular peds — slightly sticky — numerous fine roots — size of quartz pebbles increasing — smooth clear boundary to:

12 — 20

Dark yellow brown 10YR 4/4 and reddish yellow 7.5YR 6/8 sandy clay loam — breaking into weak and coarse sub-angular blocky peds — decrease in amount of roots — increase in amount of quartz rocks — some schist and sandstone fragments — some dark brown stains between cracks — gradual smooth boundary to:

20 — 30

Dark red 10R 3/6 and reddish yellow 7.5YR 6/8 clay with some dark brown 7.5YR 4/2 stains — humic matter in root traces — slightly sticky — few roots — few quartz, sandstone and schist fragments — smooth clear boundary to:

30 — 42

Red 7.5R brownish yellow 10YR 6/6 and dark brown 10YR 3/4 clay (loam) — dark brown is found only in fracture planes — soil breaking into angular peds — few roots — some dead roots — old root traces — some quartz, getting larger — some schist and sandstone fragments — gradual smooth boundary to:

42 — 50

Brownish yellow 10YR 6/6 mottled dark brown 10YR 3/4 and dusky red 7.5R 3/4 clay loam — very few roots — quartz, schist and sandstone make up about 20% of horizon — smooth clear boundary to:

50

Dark red 10R 3/6 and brownish yellow 10YR 7/8 clay loam — smooth — some dark brown stains at the sides of quartz pebbles and root traces — very few roots — the rocks are stained with iron oxide — some soft white and shine, minute specks.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.0		106	13	48	16	19	5.1	3.3	2.6	1.1	0.11	0.20	65	1.9	0.19	10.0	5
4 — 12	5.0		33	22	51	7	23	4.1	1.2	1.4	0.3	0.05	0.13	29	1.0	0.11	9.1	2
12 — 20	5.0		23	17	47	3	35	4.1	0.6	0.9	0.4	0.05	0.06	15	0.6	0.07	8.6	2
20 — 30	5.0		20	9	36	4	51	7.2	0.7	0.7	0.3	0.05	0.12	10	0.7	0.09	7.8	Trace
30 — 42	4.9		19	7	28	5	62	8.2	0.4	0.6	0.6	0.05	0.09	5				"
42 — 50	4.9		19	6	34	10	49	6.1	0.4	0.4	0.3	0.02	0.01	7				"
50	4.9		21	8	32	7	54	7.2	0.4	0.5	0.6	0.03	0.01	6				"

LOCATION: 100 yds. south west of St. Joseph Hospital
 DRAINAGE: Internal drainage free; run off moderate
 VEGETATION: Pangola grass field, sensitive plant, porters bush and broom grass.

ST. AUGUSTINE SERIES

TOPOGRAPHY: Slightly undulating terrace
 SLOPE: C
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3

Very dark grayish brown 10YR 3/2 loam — numerous fine and medium roots — weak fine and medium crumbs — root traces and few fine insect casts — smooth clear boundary to:

3 — 10

Dark brown 10YR 3/3 mottled few fine faint dark grayish brown 10YR 4/4 loam — many fine and medium roots — root traces — root channels — structure of weak and medium sub-angular blocky peds — slightly hard — smooth clear boundary to:

10 — 20

Light reddish brown 2.5YR 6/4 mottled few fine faint dark grayish brown 10YR 4/2 and few fine faint light brownish gray 2.5Y 6/2 loam — many fine roots — few dark gray 5Y 4/1 infilled tunnels — structure of weak medium sub-angular blocky peds — few fine tubular pores — slightly hard — gradual wavy boundary to:

20 — 42

Red 2.5YR 4/6 mottled few fine distinct light olive gray 5Y 6/2 loam — few fine roots — root channels — few dead roots — few black 2.5Y N2/ stains — structure of weak medium sub-angular blocky peds — few fine sandstone and quartz grains — slightly hard — gradual wavy boundary to:

42 — 60

Red 2.5YR 4/8 and olive gray 5Y 4/2 clay loam — few fine roots — dead roots — few root channels — medium sandstone and quartz rocks — structure of weak medium and coarse crumbs — few streaks of olive gray 5Y 3/2 and black 5Y 2/1 stains in red matrix — smooth clear boundary to:

60

Weak red 10R 4/3 and olive gray 5Y 5/2 loam — few fine roots — few root channels — dead roots — few fine tubular pores — few schist particles — few sandstone and quartz rocks — structure of weak medium sub-angular blocky peds — compact in pit — easy to crush when taken out — black 5Y 2/1 stains on ped faces.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	*C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3*	6.4		250	6	18	24	42	11.4	15.6	14.0	1.6	0.13	1.33	100	2.2	0.43	5.1	11
3 — 10*	6.2		142	5	23	26	47	11.3	10.8	10.5	1.1	0.04	0.45	96	2.7	0.34	7.9	4
10 — 20*	6.1		62	5	21	26	51	7.8	6.2	6.4	0.7	0.03	0.75	79	1.3	0.23	5.7	0
20 — 42	5.8		35	6	17	22	58	6.3	3.6	3.9	0.3	0.02	0.17	57				0
42 — 60	5.6		14	13	19	22	51	4.6	2.8	2.9	0.3	0.02	0.17	61				0
60	6.0		16	12	24	20	45	4.4	2.8	3.0	0.3	0.02	0.15	64				0

*High values suggest recent addition of fertilizers.

SOIL UNIT: 243

SANTA CRUZ SERIES

LOCATION: Huggins Estate on the Saddle Road
 DRAINAGE: Internal drainage free; run off low
 VEGETATION: Good stand of grape fruit, Gamelot⁽¹⁾ grass, Spanish needle, bull grass, and some small shrubs; generally thick undergrowth.

TOPOGRAPHY: Gentle slopes from hills
 SLOPE: B
 EROSION: 1

(1) Gamelote

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Dark brown 10YR 3/3 sandy loam — many fine and medium roots interlaced with each other — weak fine and medium crumbs — loose — small bits of sandstone at the surface — smooth clear boundary to:
 5 — 11 Light yellowish brown 2.5YR 6/4 loamy sand — breaking into weak coarse crumbs — smooth feel — micaceous — some flat, soft black particles — bits of sandstone — very few roots — smooth clear boundary to:
 11 — 17 Light yellowish brown 2.5Y 6/4 and reddish brown 5YR 5/3 sandy loam — small bits of sandstone — roots have decreased appreciably in this horizon — some rocks that are dark olive in colour — clear wavy boundary to:
 17 — 30 Reddish yellow 5YR 6/6 mottled olive yellow 2.5Y 6/8 gravelly loam — micaceous — soft sandstone rocks — some dark brown 10YR 3/3 soft rocks which could easily be separated into thin plates — smooth clear boundary to:
 30 — 38 Large block of yellow brown sandstone:
 38 Brownish yellow 10YR 6/6 loamy sand — small flat soft particles, black and silvery in colour — very few roots — rocks make up about 50% of this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec cond mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		114	24	59	8	14	2.6	2.6	2.6	0.6	0.05	0.09	100	1.3	0.12	10.8	6
5 — 11	5.2		28	14	64	5	18	1.5	1.4	1.1	0.2	0.03	0.11	93	0.5	0.6	8.3	2
11 — 17	5.4		13	16	60	5	22	1.3	1.0	0.9	0.2	0.04	0.01	77	0.4	0.5	8.0	Trace
17 — 30	5.3		15	19	52	7	26	1.0	0.6	0.6	0.2	0.03	0.01	60	0.4	0.5	8.0	0
30 — 38	—			SANDSTONE BOULDER														
38 +	5.8		14	21	59	5	16	1.2	0.6	0.3	0.2	0.02	0.00	50				Trace

SOIL UNIT: 53

ANGLAIS SERIES

LOCATION: Small Cocoa Estate — Spicer Road
 DRAINAGE: Free
 VEGETATION: Cocoa, bananas, coffee, ferns, razor grass and broom grass

TOPOGRAPHY Undulating hills
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark grayish brown 2.5Y 4/2 loam — breaking into weak fine crumbs — coarse and fine roots — humic matter filtered down root traces — some burnt material at the surface — abrupt smooth boundary to:
 3 — 7 Yellowish brown 10YR 5/6 and light olive brown 2.5Y 5/4 clay loam — friable — many fine roots — weak medium and coarse sub-angular peds — some grayish brown 10YR 5/2 following root traces — roots more abundant in light olive brown rather than the yellowish brown material — smooth clear boundary to:
 7 — 13 Reddish yellow 7.5YR 6/8 and pale olive 5Y 6/4 loam — many fine roots — dark greyish brown 10YR 5/2 following root traces — slightly sticky — soil breaking into coarse angular peds, which are easy to crush — smooth clear boundary to:
 13 — 31 Gray 5Y 5/1 and light reddish yellow 5YR 6/8 clay — many fine roots — structure of medium and coarse angular peds — loose — slightly moist — two iron stone concretions about 2 inches in diameter — clear wavy boundary to:
 31 — 38 Gray 7.5YR N7/ mottled red 2.5YR 5/6 silty clay loam — some sandstone and iron stone concretions — few fine roots — slightly sticky — structure of angular peds, which are easy to crush — smooth clear boundary to:
 38 — 44 Light gray 2.5Y N7/ and red 2.5YR 4/6 clay — slightly sticky — some small schist fragments — sandstone — soil breaking into weak angular peds — very few fine roots — abrupt smooth boundary to:
 44 Gray 2.5YR N5/, brown 7.5YR 5/6 and light red 2.5YR 6/8 schist and sandstone — some old root traces.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.0		105	2	28	18	42	8.3	6.1	3.3	3.1	0.23	0.45	74	2.0	0.19	10.5	5
3 — 7	4.9		25	2	31	18	45	5.6	3.0	2.2	0.7	0.12	0.21	54	0.7	0.10	7.0	2
7 — 13	4.9		16	2	35	20	49	8.5	2.4	1.8	0.9	0.11	0.35	28	0.5	0.09	5.6	1
13 — 31	4.9		14	2	34	18	49	6.3	1.9	1.0	0.4	0.10	0.60	30				Trace
31 — 38	4.8		18	1	29	16	56	7.1	1.1	0.9	0.4	0.09	0.77	15				"
38 — 44	4.7		19	1	34	18	53	7.7	2.2	0.8	0.4	0.09	1.31	29				"

LOCATION: Arima Old Road
 DRAINAGE: Imperfect
 VEGETATION: L'Epinet, coconut palm, wild coffee, wild tobacco,
 Christmas bush, bois canot and corn grass

CLEAVER SERIES

TOPOGRAPHY Low, undulating hills
 SLOPE: C
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Very dark brown 10YR 2/2 humic sandy loam — many roots interlaced with each other — weak fine and medium crumbs — loose — many small rounded quartz pebbles 1/2" in diameter with some sandstone fragments making up about 40% of horizon — smooth clear boundary to:
 3 — 9 Dark gray 10YR 4/1 mottled fine faint yellowish brown 10YR 5/6 loam — micaceous — numerous roots — weak medium crumbs — loose — quartz and sandstone rocks make up about 30% of horizon — slightly sticky — abrupt smooth boundary to:
 9 — 19 Dark brown 7.5YR 4/4 mottled few medium distinct yellowish red 5YR 5/8 clay loam — many roots — slightly sticky — structure of weak medium sub-angular peds — few quartz and sandstone rocks — smooth clear boundary to:
 19 — 32 Red 7.5R 5/6 mottled reddish brown 2.5YR 4/4 clay loam — slightly sticky — slightly plastic — breaking into weak coarse sub-angular peds — few roots — compact — weakly cemented — abrupt smooth boundary to:
 32 — 52 Red 7.5R 5/6 mottled few fine distinct light gray 5Y 7/1 clay — few roots — some root channels — sticky — slightly plastic — cemented dusky red 5R 3/3 — the few quartz rocks are also stained dusky red — soil breaking into irregular peds — few roots following crack faces — smooth clear boundary to:
 52 Dark red 5R 3/6 and light gray 5Y 7/2 clay — breaking into weak medium and coarse angular peds — fracture planes are along gray material — in the light gray material there were some distinct light greenish gray 5G 7/1 splotches — very few roots — appreciable decrease in numbers of rock fragments, but an increase in size over those in the above horizons — rocks are stained dusky red when broken.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3		5.2	61	26	33	19	25	7.2	4.3	3.2	1.8	0.13	0.07	60	3.6	0.30	12.0	16
3 — 9		4.5	48	14	40	23	30	5.8	0.4	0.3	0.2	0.10	0.02	7	3.2	0.23	13.9	10
9 — 19		4.5	30	13	41	19	34	6.1	0.2	0.0	0.0	0.05	0.02	0	1.4	0.12	11.7	1
10 — 32		4.5	22	8	27	13	58	8.8	0.0	0.0	0.0	0.06	0.02	0				Trace
32 — 52		4.2	77	10	19	13	62	7.3	0.0	0.0	0.0	0.05	0.07	0				"
52+		4.3	37	3	14	17	70	11.4	0.0	0.0	0.0	0.06	0.04	0				"

LOCATION: Old Valencia Road
DRAINAGE: Free in top soil
 Imperfect in subsoil
VEGETATION: Galba, serette, wild coffee, cocorite, aluminium plants,
 incense and moderately thick undergrowth of shrubs

PIARCO SERIES

TOPOGRAPHY: Mildly undulating
SLOPE: B
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Grayish brown 2.5Y 5/2 mottled few fine faint dull dark brown 10YR 3/3 sandy loam — many fine and medium roots — few dead root traces — weak fine granular peds — loose — slightly sticky — non-plastic — smooth clear boundary to:
 5 — 15 Light gray 7.5YR N7/ mottled few fine distinct strong brown 7.5YR 5/8 and few fine distinct reddish brown 2.5YR 4/4 sandy loam — few fine roots — few medium flutes of dark gray 5YR 4/1 humic matter — loose — slightly sticky — non-plastic — abrupt smooth boundary to:
 15 — 38 Light gray 10YR 7/1 mottled few fine distinct red 2.5YR 4/6 and few medium distinct brown 7.5 YR 5/4 loam — few fine roots — moderately compact — difficult to dig — easy to crush — slightly sticky — non-plastic — clear wavy boundary to:
 38 — 54 Light gray 5Y 7/1 mottled few fine distinct light brown 7.5YR 6/4 and few fine distinct red 2.5YR 4/6 loam — few roots — structure of weak medium sub-angular peds — weakly cemented — dead reddish brown 2.5YR 4/4 root traces — smooth clear boundary to:
 54 — 66 Light gray 7.5YR N7/ mottled few fine distinct reddish yellow 7.5YR 6/8 and few fine distinct red 2.5YR 4/8 sandy loam — very few fine roots — compact in pit — easy to crush — few dead roots — weakly cemented — slightly sticky — slightly plastic — clear irregular boundary to:
 66 Red 2.5YR 4/8, yellowish brown 10YR 5/6 and dark grayish brown 2.5Y 4/2 sandy clay loam — very few fine roots — compact — strongly cemented — structure of irregular sub-angular peds — slightly sticky — slightly plastic — water seeping into pit from sides.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	4.3		50	13	60	9	18	2.0	0.4	0.0	0.5	0.05	0.71	20	1.0	0.09	11.1	2
5 — 15	4.4		24	14	49	11	20	1.9	0.8	0.0	0.0	0.03	0.24	42	0.2	0.03	6.7	1
15 — 38	4.3		26	12	41	9	40	3.7	0.6	0.0	0.2	0.06	1.03	16				1
38 — 54	4.4		29	5	32	9	52	5.7	0.8	0.0	0.2	0.07	1.20	14				Trace
54 — 66	4.6		17	13	48	11	28	2.9	0.2	0.0	0.2	0.05	1.09	7				1
66	4.6		20	24	33	13	30	1.9	0.4	0.0	0.2	0.07	2.50	21				Trace

LOCATION: SOIL UNIT: 255
Long Stretch — opposite 23 mile post

DRAINAGE: Imperfect to impeded

VEGETATION: Galba, various shrubs, including aluminium plants.

PROFILE DESCRIPTION

VALENCIA SERIES

TOPOGRAPHY: Slope A and B, numerous hog-wallows.

SLOPE: B

EROSION: 1

Depth (ins.)

0 — 4

Dark gray 10YR 4/1 and gray 10YR 5/1 sandy clay loam — many fine and medium roots — few fine tubular pores — structure of weak medium sub-angular blocky peds — slightly sticky — slightly plastic — abrupt smooth boundary to:

4 — 12

Light gray 10YR 7/1 and dark yellowish brown 10YR 4/4 mottled few fine faint brownish yellow 10YR 6/6 sandy loam — few fine roots — structure of fine medium irregular sub-angular peds — few fine tubular pores — few dead root traces — loose — smooth clear boundary to:

12 — 24

Light gray 5Y 7/1 mottled few fine faint brownish yellow 10YR 6/6 sandy loam — loose — few fine roots — few dead root traces — structure of weak fine irregular sub-angular peds — few fine tubular pores — slightly sticky — non-plastic — smooth clear boundary to:

24 — 48

Light olive gray 5Y 6/2, light yellowish brown 10YR 6/4 and red 2.5YR 4/6 sandy loam — breaking into medium sub-angular peds — few fine roots — weakly cemented — compact — slightly sticky — non-plastic — clear wavy boundary to:

48 — 56

Pale olive 5Y 6/4, light yellowish brown 10YR 6/4 and dark red 2.5YR 3/6 sandy clay loam — breaking into moderate sub-angular peds — compact — firm — strongly cemented — developing into indurated material — sticky — slightly plastic — few small quartz rocks about 1/5 inch in diameter — smooth clear boundary to:

56

Red 2.5YR 4/8, pale yellow 5Y 7/4 and pale olive 5Y 6/4 loam — breaking into moderate medium sub-angular peds — compact — firm — cemented — slightly indurated — slightly sticky — slightly plastic — few small quartz pebbles.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	4.5		65	11	47	17	20	3.0	0.2	0.0	0.9	0.08	0.42	7	1.4	0.10	14.0	3
4 — 12	4.9		25	10	48	17	22	1.8	0.2	0.0	0.5	0.04	0.33	11	0.7	0.05	14.0	2
12 — 24	4.8		18	10	39	15	30	2.2	0.0	0.0	0.9	0.04	1.20	0				2
24 — 48	4.9		16	8	41	11	39	4.4	0.0	0.0	1.3	0.04	0.05	0				1
48 — 56	4.9		15	12	51	13	30	3.4	0.0	1.2	1.4	0.08	0.48	0				2
56	4.9		15	20	16	11	56	5.0	0.0	0.0	2.2	0.12	2.16	0				2

LOCATION: SOIL UNIT: 59
Long Stretch

DRAINAGE: Impeded
VEGETATION: Serrette, galba, timite, cocorite and various hard-wood shrubs

LONG STRETCH SERIES
TOPOGRAPHY: Flat with hog-wallows
A and B slopes
SLOPE: A
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 4

Light gray 10YR 6/1 silty loam — thin root mat — slightly sticky — coarse weak crumbs — friable — abrupt smooth boundary to:

4 — 10

Light gray 7.5YR N7/ mottled few, distinct, medium reddish yellow 7.5YR 6/6 with dark gray 7.5YR N4/ stains, silty clay loam — humic matter filtering down root traces — many thin roots — structure of sub-angular peds — slightly sticky — slightly plastic — clear wavy boundary to:

10 — 18

Light gray 10YR 7/1 mottled few fine distinct dark brown 7.5YR 4/4 sandy clay loam — many thin roots — structure of irregular sub-angular peds — roots following fracture planes — slightly sticky — slightly plastic — compact — firm — small flutes that are dark brown 7.5YR 2/2 — smooth clear boundary to:

18 — 36

Light gray 7.5YR N7/ mottled few, distinct coarse brown 7.5YR 5/6 sandy clay loam — few thin roots — some dead root traces — slightly plastic — compact — firm — smooth clear boundary to:

36 — 48

Light gray 2.5Y N7/ and yellowish brown 10YR 5/6 mottled few distinct medium dark red 7.5YR 3/6 silty clay — few roots — some dead roots — weakly cemented by red 10R 4/6 concretions — smooth clear boundary to:

48 — 56

Light bluish gray 2.5Y N7/ mottled few, prominent coarse reddish yellow 7.5YR 6/8, and few coarse distinct dusky red 7.5R 3/4 silty clay — very few thin roots — some dead roots — slightly sticky — slightly plastic — structure of moderate irregular sub-angular peds — compact — weakly cemented — some reddish yellow markings — smooth clear boundary to:

56

Light gray 2.5Y 7/2 mottled few coarse distinct reddish yellow 7.5YR 7/6 and few, coarse, prominent dark red 10R 3/6 silty clay loam — very few roots — weakly cemented — breaking into irregular fragments — compact — firm — a definite increase in the red colour.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	4.2		69	1	45	30	24	3.0	0.0	0.0	0.0	0.00	0.04	0	1.7	0.13	13.1	7
4 — 10	4.1		42	1	42	32	28	2.6	0.0	0.0	0.0	0.00	0.01	0	0.9	0.08	11.3	2
10 — 18	4.4		23	0	41	26	31	1.1	0.0	0.0	0.0	0.00	0.01	0	0.5	0.05	10.0	1
18 — 36	4.4		22	0	34	24	41	4.7	0.0	0.0	0.0	0.00	0.03	0				1
36 — 48	4.3		34	0	15	14	69	8.5	0.0	0.0	0.0	0.01	0.04	0				1
48 — 56	4.4		27	0	19	17	69	8.1	0.0	0.0	0.0	0.05	0.10	0				Trace
56	4.4		28	0	24	15	66	5.0	0.0	0.0	0.0	0.05	0.10	0				"

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3

Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:

3 — 9

Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:

9 — 16

Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:

16 — 28

Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:

28 — 48

Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:

48

Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
DRAINAGE: Free
VEGETATION: Crappo, sardine bush, olivier, and many hard-wood shrubs

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION: Maraval-Fatima Hill
DRAINAGE: External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION: Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

[illegible]

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:

3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:

9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:

24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:

42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:

60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION: Maraval-Fatima Hill
DRAINAGE: External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION: Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

[illegible]

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION: Maraval-Fatima Hill
DRAINAGE: External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION: Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

[illegible]

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 181

MATELOT SERIES

LOCATION:
DRAINAGE:
VEGETATION:

Blanchisseuse near 20¼ mile post
Internal drainage free, encouraged by vertical and lateral movement along the schist: run off medium to rapid
Coconut palms, wild coffee, serrette and small shrubs — it appeared that fire had recently burnt the surrounding bush.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9 Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30 Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44 Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44 Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil				% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P		
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg		K	Na			C	N
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2
																		"

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
DRAINAGE: Free
VEGETATION: Crappo, sardine bush, olivier, and many hard-wood shrubs

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

- 0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
- 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
- 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
- 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
- 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
- 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil					m.e. per 100 gm. Oven dry soil					% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60 +	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

SOIL UNIT: 82/L

DIEGO MARTIN SERIES

Maraval-Fatima Hill

TOPOGRAPHY: Hilly

External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.

SLOPE: D

Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

EROSION: 1

LOCATION:

DRAINAGE:

VEGETATION:

ROFILE DESCRIPTION

Depth (ins.)

0 — 3

Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:

3 — 7

Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:

7 — 13

Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:

13 — 32

Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:

32 — 38

Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:

38 — 54

Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:

54

Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil					% Base Sat.		% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na	C	N			
0 — 3	7.6	0.6	368	15	50	27	20							3.9	0.46	8.5	—	
3 — 7	7.7	9.4	224	16	52	24	22							2.1	0.28	7.5	—	
7 — 13	8.1	37.4	112	29	14	16	21										—	
13 — 32	8.3	47.4	96	39	42	15	16										—	
32 — 38	8.2	60.4	80	53	31	13	13										—	
38 — 54	8.4	64.2	88	41	43	16	13										—	

Free lime in profile

SOIL UNIT: 81

MARACAS SERIES

LOCATION:

Carenage

DRAINAGE:

Internal drainage free; run off medium to rapid

VEGETATION:

Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope

SLOPE: D

EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3

Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:

3 — 9

Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:

9 — 16

Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:

16 — 28

Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:

28 — 48

Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:

48

Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	1.5	0.17	8.8	3	
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	1.3	0.15	8.7	2	
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	1.2	0.13	9.2	Trace	
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	1.0	0.11	9.1	0	
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00				1	
48 +	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04				1	

SOIL UNIT: 181

MATELOT SERIES

LOCATION:
DRAINAGE:
VEGETATION:

Blanchisseuse near 20¼ mile post
Internal drainage free, encouraged by vertical and lateral movement along the schist: run off medium to rapid
Coconut palms, wild coffee, serrette and small shrubs — it appeared that fire had recently burnt the surrounding bush.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9 Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30 Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44 Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44 Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil				% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P		
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg		K	Na			C	N
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2
																		"

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:

3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:

9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:

24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:

42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:

60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION: Maraval-Fatima Hill
DRAINAGE: External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION: Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

[illegible]

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:

3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:

9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:

24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:

42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:

60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION: Maraval-Fatima Hill
DRAINAGE: External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION: Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	7.6	0.6	368	15	50	27	20								3.9	0.46	8.5	—
3 — 7	7.7	9.4	224	16	52	24	22								2.1	0.28	7.5	—
7 — 13	8.1	37.4	112	29	14	16	21											—
13 — 32	8.3	47.4	96	39	42	15	16	Free lime in profile									—	
32 — 38	8.2	60.4	80	53	31	13	13											—
38 — 54	8.4	64.2	88	41	43	16	13											—

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
DRAINAGE: Free
VEGETATION: Crappo, sardine bush, olivier, and many hard-wood shrubs

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION: Maraval-Fatima Hill
DRAINAGE: External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION: Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

[illegible]

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
DRAINAGE: Free
VEGETATION: Crappo, sardine bush, olivier, and many hard-wood shrubs

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION: Maraval-Fatima Hill
DRAINAGE: External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION: Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

[illegible]

LOCATION: Government forest, La Pastora Demonstration Station —
 pit on the edge of the cocoa plantation
 DRAINAGE: Free
 VEGETATION: Forest trees, cock spur, balisier and cigarette bush

LA PASTORA SERIES

TOPOGRAPHY: Hilly (pit on moderate
 slope)
 SLOPE: C
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 4 Very dark brown 10YR 3/3 clay loam — fine and medium crumbs — numerous roots — slightly sticky — smooth clear boundary to:
 4 — 14 Reddish brown 5YR 4/3 clay loam — slightly sticky — medium and fine sub-angular blocky peds — many roots following fracture planes — roots along crack faces — abrupt wavy boundary to:
 14 — 26 Dark reddish brown 5YR 3/4 and brown 7.5YR 5/4 clay — few large and thin roots — some small rock fragments — soil breaking into medium prismatic peds — slightly sticky — slightly plastic — smooth clear boundary to:
 26 — 38 Reddish brown 2.5YR 4/4 clay loam — few roots — some quartz and sandstone fragments — slightly sticky — firm — structure of weak irregular blocky peds — smooth clear boundary to:
 38 — 44 Brown 7.5YR 5/2 clay loam — breaking into weak medium and coarse sub-angular peds — slightly sticky — black 2.5YR N3/ stains on ped faces — many rock fragments:
 44 Parent material — large dark coloured calcareous rocks.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	6.7	0.1	330	18	30	14	40	12.3	14.1	10.9	3.4	0.01	0.11	100	1.6	0.31	5.2	11
4 — 14	7.1		98	17	33	11	41	7.1	7.7	6.7	0.9	0.00	0.01	100	0.9	0.16	5.6	4
14 — 26	7.3		47	14	27	12	53	5.0	4.5	4.0	0.3	0.00	0.02	90	0.5	0.12	4.2	1
26 — 38	6.9		30	16	33	7	44	3.2	3.2	2.8	0.2	0.00	0.00	100				1
38 — 44	7.2		36	13	34	14	39	4.4	4.4	4.2	0.3	0.00	0.07	100				36

LOCATION: Near Quarry on Saddle Road: chive garden nearby
DRAINAGE: Internal drainage free, run-off rapid
VEGETATION: Forest — mahogany, salt fish wood, (legume), green heart and various small trees.

MARAVAL SERIES
TOPOGRAPHY: Hilly
SLOPE: C
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2 Dark brown 7.5YR 3/2 silty loam — numerous thin roots — fine and medium crumbs — aggregates easy to crush — smooth clear boundary to:
 2 — 5 Reddish brown 5YR 4/4 to yellowish red 5YR 4/8 sandy loam — breaking into weak medium and coarse crumbs — numerous thin roots — easy to crush — some very small schist particles — smooth to the feel — smooth clear boundary to:
 5 — 11 Dark reddish brown 2.5YR 2/5 silty clay loam — breaking into weak fine and medium sub-angular blocky peds — few roots — schistose fragments — abrupt smooth boundary to:
 11 — 27 Yellowish red 5YR 4/8 silt clay loam — few fine roots — smooth feel due to schist particles — smooth clear boundary to:
 27 — 44 Yellowish red 5YR 4/6 mottled reddish brown 2.5YR 4/4 silty clay loam — very few fine roots — few dead roots — small dark coloured schist particles — clear wavy boundary to:
 44 — 59 Yellowish brown 10YR 5/6 mottled pale brown 10YR 7/3 loam — increase in sand content — very few roots — dead roots — thin gravel layer composed of schist and quartz and a few bits of sandstone — gradual wavy boundary to:
 59 Very pale yellow 10YR 7/3 mottled white 10YR 8/2 and black 2.5YR N2/ — weathered micaceous schist — bits of limestone — some roots — a large block of limestone at the bottom.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.5		517	10	23	26	34	22.9	29.0	22.4	7.7	0.40	0.52	100	2.7	0.54	5.0	17
2 — 5	6.5		333	12	24	24	38	15.3	16.2	12.5	4.6	0.08	0.18	100	2.5	0.35	7.1	6
5 — 11	6.7		55	13	27	21	42	7.6	6.0	5.0	1.4	0.04	0.09	79	0.9	0.18	5.0	6
11 — 27	6.6		40	14	30	12	50	5.1	4.0	3.4	0.9	0.00	0.04	78	0.5	0.11	4.5	Trace
27 — 44	6.8		25	15	28	16	44	4.1	2.9	2.5	0.4	0.00	0.00	72				1
44 — 59	6.8		27	16	26	16	43	3.6	2.5	2.3	0.2	0.01	0.04	69				1
59 +	7.2		36	18	34	26	26	3.3	3.8	3.5	0.2	0.01	0.13	100				31

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:
 3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:
 9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:
 24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:
 42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:
 60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

SOIL UNIT: 261

ARENA SAND

LOCATION: Siparia Old Road, approximately 1 mile from Siparia Erin Road

DRAINAGE: Free to excessive

VEGETATION: Secondary forest — black sage, Christmas bush, bois conot and laurier.

TOPOGRAPHY: Hilly.

SLOPE: C

EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 5 Black 10YR 2/1 humic sand — numerous coarse and fine roots — weak fine granular structure — loose — few colourless quartz grains — abrupt clear boundary to:

5 — 23 Dark brown 5.5YR 4/4, few white 10YR 8/1 specks loamy sand — few fine and medium roots — sand grains coated with humic matter — weak fine granular structure — loose — smooth clear boundary to:

23 — 36 Dark yellow brown 10YR 4/4 mottled few fine distinct light gray 10YR 7/1 loamy sand — few medium and fine roots — loose — humic matter filtered down from the top — abrupt smooth boundary to:

36 — 50 Very pale brown 10YR 7/3 loamy sand — very few roots — few dead root traces — loose — smooth clear boundary to:

50 Very pale brown 10YR 8/3 mottled few medium distinct brown 10YR 5/4 loamy sand — one root — loose — slightly compact but easy to crush — few fine dead root traces — some of the channels filled with darker material.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 5	5.7		138	56	32	7	11	4.7	5.3	5.2	0.8	0.06	0.13	100	1.1	0.12	9.2	5
5 — 23	5.9		27	65	24	3	15	2.1	1.4	0.9	0.4	0.04	0.00	67	0.3	0.04	7.5	4
23 — 36	5.8		21	75	20	3	15	2.6	0.8	0.2	0.4	0.02	0.00	31				3
36 — 50	5.9		14	64	26	3	11	1.4	0.5	0.2	0.4	0.02	0.00	36				3
50	5.8		16	75	20	3	11	1.1	0.4	0.1	0.4	0.00	0.00	36				3

SOIL UNIT: 177

TALPARO SERIES

LOCATION: San Francique Road
 DRAINAGE: Impeded
 VEGETATION: Recently cleared secondary forest

TOPOGRAPHY: Hilly.
 SLOPE: B
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 2

Very dark gray 10YR 3/1 clay — numerous fine and medium roots — fine medium and coarse crumbs — fine and medium tubular pores — slightly sticky — slightly plastic — few worm casts — smooth clear boundary to:

2 — 10

Yellowish brown 10YR 5/4 and dark grayish brown 10YR 4/2 clay — many fine and medium roots — weak coarse angular and sub-angular peds — very dark gray 10YR 3/1 skins on ped faces — few root traces — slightly sticky — slightly plastic — compact — firm — abrupt smooth boundary to:

10 — 22

Greenish yellow 5GY 6/1 mottled few medium distinct red 10YR 4/8 clay — many fine and medium roots — few dead roots — few root channels — weak coarse sub-angular blocky peds — skins of dark gray 10YR 4/1 on ped faces — sticky — plastic — compact — firm — smooth clear boundary to:

22 — 40

Greenish yellow 5GY 5/1 and red 7.5R 4/6 clay — many fine and medium roots — dead roots — few root channels — sticky — plastic — firm — compact — gradual irregular boundary to:

40 — 60

Greenish yellow 5GY 7/1 and strong brown 7.5YR 5/8 clay — few fine roots — few dead roots — black 7.5YR N2/ manganese dioxide stains — live roots along crack faces — coarse platelike structure — breaking into weak medium angular peds — smooth clear boundary to:

60

Greenish yellow 5GY 5/1 mottled few medium distinct yellowish brown 10YR 5/6 clay shale — very few fine roots — few dead roots — coarse and medium plates — nests of gypsum — manganese dioxide stains along planes — slightly sticky — slightly plastic — compact — firm.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 2	6.1		106	4	18	15	65	34.4	36.3	18.7	16.1	2.21	0.88	100	6.2	0.61	10.2	50
2 — 10	5.4		37	3	8	10	82	29.8	24.0	8.8	13.5	1.64	1.43	81	2.6	0.28	9.3	32
10 — 22	5.0		58	0	1	4	97	35.4	24.2	5.6	17.4	1.09	2.68	68				21
22 — 40	4.3		56	0	1	4	96	34.5	23.1	3.1	19.1	0.82	6.27*	67				9
40 — 60	4.8		119	3	2	10	94	33.6	29.7	5.7	26.3	0.65	1.70*	88				61
60	5.9		325	6	2	3	90	Gypsum present										

SOIL UNIT: 81

MARACAS SERIES

LOCATION: Carenage
 DRAINAGE: Internal drainage free; run off medium to rapid
 VEGETATION: Black sage, bois canot, wild coffee and a few coconut palms

TOPOGRAPHY: Hill with steep slope
 SLOPE: D
 EROSION: 2

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 7.5YR 3/2 loam — loose fine and medium crumbs — some small bits of quartz and sandstone — numerous roots — abrupt smooth boundary to:
 3 — 9 Yellowish red 5YR 5/6 loam — breaking into weak medium and coarse crumbs — many roots — some small bits of gravel comprising of quartz and sandstone with some schist — smooth clear boundary to:
 9 — 16 Light red 2.5YR 6/6 and yellow brown 10YR 5/6 loam — silty feel due to schistose material — small, thin healthy roots — old root traces — fragments of schist, sandstone and quartz — gradual wavy boundary to:
 16 — 28 Red 2.5YR 6/6 mottled light brown 7.5YR 6/4 loam — soil has a silty feel due to schistose material — roots — old root traces — gradual smooth boundary to:
 28 — 48 Weak red 2.5YR 4/2 mottled yellow brown 10YR 5/6 loam — small fragments of schist with some quartz and sandstone pebbles — a small layer of gravel at the bottom of this horizon — smooth clear boundary to:
 48 Light red 2.5YR 6/8 and strong brown 7.5YR 5/6 loam — bits of rocks which make up about 25% of this horizon — very few roots.

ANALYSE

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	5.9		31	18	40	25	21	5.0	3.3	3.1	0.4	0.05	0.24	66	1.5	0.17	8.8	3
3 — 9	6.0		27	12	44	19	20	3.4	2.9	2.4	0.6	0.03	0.15	85	1.3	0.15	8.7	2
9 — 16	5.8		19	18	44	27	21	3.8	3.2	2.3	0.6	0.01	0.02	84	1.2	0.13	9.2	Trace
16 — 28	5.4		13	10	29	18	39	4.5	2.9	2.2	0.9	0.00	0.00	64	1.0	0.11	9.1	0
28 — 48	5.3		18	9	36	17	36	3.1	1.6	1.1	0.8	0.01	0.00	52				1
48+	5.3		21	9	37	17	36	3.5	1.2	1.1	0.7	0.01	0.04	34				1

MATELOT SERIES

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

82

Depth (ins.)	Description
0 — 4	Black 10YR 2/1 humic sandy loam — fine and medium crumbs — some gritty material and small quartz particles — slightly sticky — numerous roots — abrupt smooth boundary to:
4 — 9	Light yellow brown 10YR 6/4 sandy clay loam — breaking into weak fine and medium sub-angular peds — slightly sticky — one small fairly soft sandstone — two small iron concretions — many fine roots — smooth clear boundary to:
9 — 30	Very pale brown 10YR 7/3 loam — breaking into medium sub-angular peds — many large and small roots — some humic matter along root traces — slightly sticky — slightly plastic — smooth clear boundary to:
30 — 44	Brown 7.5YR 5/2 silty loam — silty texture may be due to schist and sandstone — when crushed is much lighter in colour — clear wavy boundary to:
44	Yellowish brown 10YR 5/6 and reddish yellow 5YR 4/8 silty clay loam — many thin roots — slightly sticky — 75% of rocks are sandstone — some quartz and bits of schist.

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10-6	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	5.4		166	25	27	22	28	10.6	9.2	7.1	2.2	0.29	0.68	87	2.7	0.19	14.4	6
4 — 9	6.1		98	29	33	13	30	5.3	6.1	5.1	1.3	0.06	0.15	100	1.7	0.12	14.2	Trace
9 — 30	5.3		41	29	33	17	32	4.4	1.4	0.8	0.7	0.00	0.00	32	1.1	0.10	11.0	"
30 — 44	5.1		21	30	24	15	33	3.8	1.0	0.5	0.7	0.04	0.04	26				"
44 +	5.2		27	31	26	17	30	2.8	0.6	0.4	0.1	0.16	0.22	21				2 "

SOIL UNIT: 281

SPRING HILL SERIES

LOCATION: Government Forest, Blanchisseuse Road
 DRAINAGE: Free
 VEGETATION: Crappo, sardine bush, olivier, and many
 hard-wood shrubs

TOPOGRAPHY: Hilly
 SLOPE: D
 EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark brown 10YR 3/3 humic loam — slightly sticky — many roots intertwined with each other — fine medium crumbs — smooth clear boundary to:

3 — 9 Yellowish brown 10YR 5/6 clay loam — slightly sticky — some dark brown 10YR 3/3 humic matter filtered from above — numerous large and small roots — fine sub-angular blocky peds — smooth clear boundary to:

9 — 24 Light yellow brown 10YR 6/4 and olive 5Y 5/4 clay — slightly sticky — small white shiny particles, probably mica — many roots — few dead ones — smooth clear boundary to:

24 — 42 Olive yellow 2.5Y 6/8 mottled yellowish red 5YR 5/8 silty clay loam — many roots — some dead root traces — slightly sticky — soil breaking into angular peds — roots following fracture planes — the yellowish red colour is abundant, forming a skin on peds — smooth clear boundary to:

42 — 60 Pale olive 5Y 6/3 and red 2.5YR 5/6 silt clay loam — very few roots — some fragments of soft red sandstone, mostly plate-like — roots between plates — irregular gradual boundary to:

60 Light olive brown 2.5Y 5/6 mottled dark reddish brown 5YR 3/4 clay loam — reddish colour is in fracture planes — no roots or stones in this horizon.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	4.7		102	15	28	16	34	11.7	1.5	0.5	1.3	0.21	0.40	13	3.2	0.27	12.0	7
3 — 9	4.9		30	5	31	18	42	6.0	0.2	0.1	0.4	0.06	0.07	33	1.3	0.15	8.6	2
9 — 24	5.0		29	7	35	20	43	5.8	0.2	0.1	0.3	0.03	0.05	34	1.2	0.13	9.2	2
24 — 42	4.7		22	5	21	16	50	4.2	0.0	0.0	0.2	0.01	0.04	0				Trace
42 — 60	4.7		22	4	25	26	47	4.1	0.0	0.0	0.2	0.02	0.00	0				"
60+	4.7		23	5	29	14	51	3.5	0.0	0.1	0.2	0.01	0.03	0				"

DIEGO MARTIN SERIES

LOCATION:	Maraval-Fatima Hill
DRAINAGE:	External drainage rapid. The top soil in this series is not severely eroded and it also has a good crumb structure as a consequence of healthy vegetative cover.
VEGETATION:	Poui, bois flot, wild coffee, thick undergrowth of shrubs and grasses.

TOPOGRAPHY: Hilly
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3	Drak grayish brown 2.5Y 4/2 sandy loam — numerous roots — fine and medium crumbs — small bits of rocks — calcareous (L1) — smooth clear boundary to:
3 — 7	Dark brown 10YR 3/3 sandy loam — medium and coarse crumbs — few roots — small bits of calcareous rocks which are soft and easy to crush — rocks are laminated and easy to split into flat plates (L2) — abrupt smooth boundary to:
7 — 13	Brownish yellow 10YR 6/5 sandy loam — weak medium sub-angular blocky peds — small fragments of limestone schist particles — few roots — the soil has a silty feel, which is probably due to schistose material (L4) — clear wavy boundary to:
13 — 32	Light yellowish brown 10YR 6/4 loam — breaking into weak coarse sub-angular blocky peds — (L5) — many fragments of soft calcareous rocks — definite decrease in number of roots — soil smooth to the touch — clear irregular boundary to:
32 — 38	Very pale brown 10YR 3/6 loam — soft calcareous rocks and some schistose material — violently calcareous — no roots seen in this horizon — abrupt clear boundary to:
38 — 54	Yellow 10YR 7/6 mottled very pale brown 10YR 8/4 loam — no roots — bits of schistose material — calcareous — soft limestone fragments — schist and limestone make up about 75% of horizon — (L5) — smooth clear boundary to:
54	Parent material — soft limestone fragments and schistose material — calcareous — no roots.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	7.6	0.6	368	15	50	27	20								3.9	0.46	8.5	—
3 — 7	7.7	9.4	224	16	52	24	22								2.1	0.28	7.5	—
7 — 13	8.1	37.4	112	29	14	16	21											—
13 — 32	8.3	47.4	96	39	42	15	16	Free lime in profile										—
32 — 38	8.2	60.4	80	53	31	13	13											—
38 — 54	8.4	64.2	88	41	43	16	13											—

SOIL UNIT: 382/L

PLATANAL SERIES

LOCATION: Heights of Aripo
DRAINAGE: Free
VEGETATION: Abandoned cocoa estate — balisier, cigarette bush, few
immortelle and few cocoa trees

RELIEF: Hilly with some mild slopes
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)
0 — 4 Dark brown 10YR 3/3 clay loam — fine and medium crumbs — numerous roots — loose — slightly sticky — abrupt smooth boundary to:
4 — 14 Brown 10YR 4/3 loam — loose — slightly sticky — many roots — weak irregular sub-angular peds — some quartz pebbles 1/2" in diameter — some limestone chips — clear gradual boundary to:
14 — 30 Strong brown 10YR 5/3 clay loam — slightly sticky — breaking into coarse sub-angular blocky peds — many loose roots — limestone block at the side of pit — some quartz grit — smooth clear boundary to:
30 — 54 Yellowish brown 10YR 5/6 mottled few fine distinct dark brown 7.5YR 4/4 clay — slightly sticky — breaking into weak coarse angular peds — roots generally following crack faces — soil at the side of limestone slightly calcareous — gradual smooth boundary to:
54 Yellowish brown 10YR 5/6 clay — many roots — root traces infilled with dark brown 10YR 3/3 humic matter — very few quartz and limestone chips — non-sticky — structure of coarse angular peds — roots following crack faces — limestone rock at the bottom of pit.

87

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	*C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 4	6.6		1218	11	24	20	49	14.6	37.9	35.2	2.7	0.27	0.96	(1)	8.1	0.93	8.7	23
4 — 14	7.1		113	11	28	19	53	7.4	10.0	10.0	0.8	0.07	0.38	100	2.3	0.29	7.9	2
14 — 30	7.6		99	9	25	14	59	6.3	8.0	8.0	0.4	0.07	0.52	100				Trace
30 — 54	7.6		49	9	24	14	59	6.3	8.2	8.2	0.3	0.04	0.59	100				65
54+	8.2		71	9	25	14	61	13.1	11.0	11.0	0.6	0.04	0.54	84				73

(1) O.M. is organic matter

SOIL UNIT: 91

SANS SOUCI SERIES

LOCATION: Peasant holding, Sans Souci Village
DRAINAGE: Imperfect
VEGETATION: Peasant garden — cassava, avocados, few coconuts, few balata and bois flot trees, razor grass

TOPOGRAPHY: Rolling hills
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 3 Dark grayish brown 10YR 4/2 clay loam — weak fine and medium crumbs — slightly sticky — numerous roots — crumbs easy to crush — crushed material lighter in colour — roots follow interfaces of crumbs — smooth clear boundary to:

3 — 10 Dark yellow brown 10YR 4/4 clay loam — black and olive yellow, soft weathered rocks — soil breaking into weak fine and medium angular crumbs — many roots — some soft weathered bits of rocks — slightly sticky — wavy clear boundary to:

10 — 24 Yellowish red 5YR 5/8 mottled grayish brown 2.5Y 5/2 clay — medium and coarse angular crumbs — sticky — slightly plastic — larger crumbs than the top — many roots — some soft weathered rocks — dark stains along laminations, probably carbon — smooth clear boundary to:

24 — 40 Very pale brown 10YR 8/3 mottled yellow red 5YR 5/8 clay — sticky — some roots — increase in partly weathered rocks with black stains (probably carbon) in fracture planes — sub-angular blocky peds — firm — smooth clear boundary to:

40 — 60 Reddish brown 5YR 5/4 and pinkish gray 7.5YR 7/2 loam — black 5YR 2/1 seams in fracture planes — few roots — soil dry and slightly hard — some weathered and un-weathered material breaking into large irregular fragments — smooth clear boundary to:

60 Red 2.5YR 3/6 and yellowish brown 10YR 5/4 loam — colour in fracture planes is dark reddish brown 2.5YR 3/4 — black 2.5YR N2/ parent rock — few roots — the entire horizon comprises of partly weathered and un-weathered parent rock.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	* C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 3	6.2		130	3	30	21	33	33.9	46.8	21.2	28.4	0.08	1.12	100	2.9	0.27	10.7	5
3 — 10	6.0		60	3	13	14	37	34.3	43.0	16.4	27.8	0.02	1.70	100	1.1	0.10	11.0	2
10 — 24	6.2		55	3	19	10	33	34.2	45.1	15.6	31.1	0.01	2.44	100	0.3	0.04	7.5	1
24 — 40	6.5		63	11	21	10	35	35.6	44.7	15.9	30.9	0.02	3.10	100				2
40 — 60	6.3		55	11	25	9	39	33.6	46.4	15.3	29.8	0.02	2.96	100				42
60+	6.3		58	3	31	12	18	32.2	47.9	16.7	30.9	0.02	2.84	100				148

The mechanical analyses results show the difficulty experienced in recovery of the various fractions.

LOCATION:
DRAINAGE:
VEGETATION:

SOIL UNIT: 94/L
Basin Galape Road, Toco
Imperfect
Cocoa, bananas, cigarette bush, stinging nettles and various types of grasses

TOCO SERIES

TOPOGRAPHY: Hilly, pit in the middle of D slope
SLOPE: D
EROSION: 1

PROFILE DESCRIPTION

Depth (ins.)

0 — 6

Dark olive gray 5Y 3/2 clay — numerous fine and medium roots — moderate coarse crumbs — bits of calcareous and carbonaceous material — worms and worm casts — slightly sticky — slightly plastic — slightly calcareous smooth clear boundary to:

6 — 12

Gray 5Y 5/1 mottled strong brown 7.5YR 5/8 clay — many fine medium and coarse roots — moderate medium sub-angular peds — breaking into weak coarse crumbs — few worms and worm casts — bits of hard carbonaceous and calcareous material — slightly sticky — slightly plastic — slightly calcareous — dark grayish brown 10YR 4/2 stains on ped faces — abrupt smooth boundary to:

12 — 24

Strong brown 7.5YR 5/6 and olive brown 2.5YR 4/4 clay — many medium roots — structure of medium and coarse prismatic peds — fracture planes smooth — few bits of carbonaceous schist — slightly sticky — slightly plastic — irregular smooth boundary to:

24 — 36

Olive gray 5Y 4/2 mottled dark brown 7.5YR 4/2 clay loam — fine and medium roots — few dead roots and root channels — structure of weak medium prismatic peds — few bits of laminated carbonaceous schist material — slightly sticky — slightly plastic — smooth clear boundary to:

36 — 42

Olive gray 5Y 5/2 and pale olive 5Y 6/3 clay — few medium and fine roots — structure of weak irregular sub-angular blocky peds — bits of carbonaceous schist and quartz — strongly calcareous — slightly sticky — slightly plastic — smooth clear boundary to:

42

Black 5Y 2/1 and dark gray 5Y 4/1 mottled few medium distinct yellow brown 10YR 5/6 clay loam — few fine roots — mixture of soil and carbonaceous material — slightly sticky — slightly plastic — violently calcareous.

ANALYSES

Depth (ins.)	pH	% Free CaCO ₃	Elec. cond. mhos X 10 ⁻⁶	Mechanical Analysis % Oven dry soil				m.e. per 100 gm. Oven dry soil						% Base Sat.	% Oven dry soil		C/N Ratio	p.p.m. P
				Crse. Sand	Fine Sand	Silt	Clay	C.E.C.	T.E.B.	Ca	Mg	K	Na		C	N		
0 — 6	7.7	4.8	247	20	24	12	41								1.2	0.17	7.1	—
6 — 12	7.4	2.3	182	20	24	12	42								0.7	0.14	5.0	—
12 — 24	6.7	—	264	12	22	12	43								1.2	0.18	6.7	—
24 — 36	6.7	—	148	28	25	12	35	Free Calcium Carbonate in profile							0.7	0.15	4.7	—
36 — 42	7.9	3.3	214	22	21	16	41											—
42	8.0	10.9	264	8	30	18	41											—

APPENDIX II
ACREAGES OF SOIL TYPES DIVIDED INTO SLOPE CLASSES

SLOPE CLASSES							TOTAL ACRES
SOIL SERIES	A	B	C	D	E	F	
1	75	75	—	—	—	—	150
15	150	—	—	—	—	—	150
123	675	1850	—	—	—	—	2525
223	1650	2300	—	—	—	—	6950
323	75	400	275	—	—	—	750
423	25	—	—	—	—	—	25
24/L	50	50	—	—	—	—	100
233	700	1425	400	—	—	—	2525
39	50	25	—	—	—	—	75
41	—	—	1000	1150	1750	—	3900
141	75	400	250	400	—	—	1125
241	—	50	—	50	—	—	100
43	—	475	400	1575	—	—	2450
143	—	2400	1475	950	—	—	4825
243	50	175	100	750	—	—	1075
53	—	275	375	875	4950	5850	12325
153	—	800	4925	—	—	—	5725
253	Less than 25 acres		—	—	—	—	—
55	—	—	3350	—	—	—	3350
255	—	125	4425	—	—	—	4550
59	—	—	975	—	—	—	975
261	—	100	75	—	—	—	175
177	—	323	350	125	50	675	1525
178	—	—	25	—	—	—	25
81	—	—	1125	1575	16425	91725	111850
181	—	—	525	1675	10075	61050	73325
281	—	—	50	—	5800	—	5850
82/L	—	—	50	250	1275	1425	3000
182/L	—	—	—	50	875	1350	2275
282/L	—	—	75	225	1275	1500	3075
382/L	—	—	—	75	3150	9025	12250
91	—	—	—	25	175	1375	1575
94/L	—	—	50	100	1600	1650	3400
	6575	11250	20275	10850	47400	175625	271975